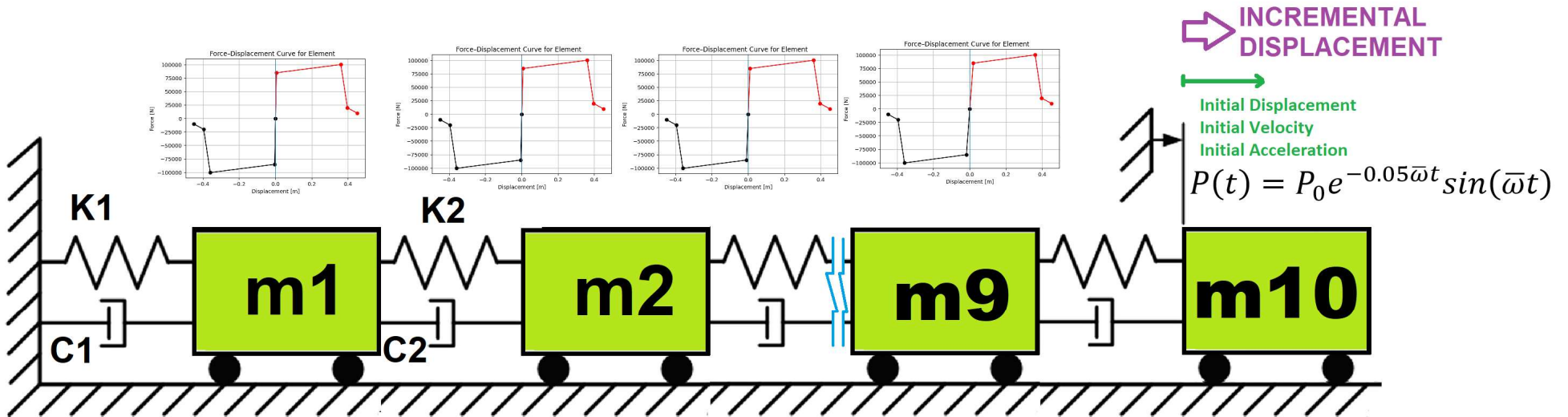


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

FRAGILITY ANALYSIS BASED ON ACCELERATION, STRUCTURAL DUCTILITY DAMAGE INDEX, ENERGY DISSIPATION CAPACITY INDEX, EQUIVALENT VISCOUS DAMPING RATIO WITH NONLINEAR DYNAMIC ANALYSIS OF A MULTI-DEGREE-OF-FREEDOM (MDOF) SYSTEM UTILIZING 200 GROUND MOTIONS IN OPENSEES

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

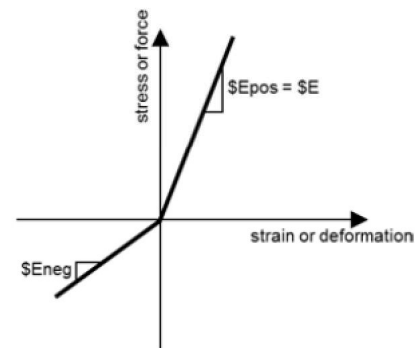
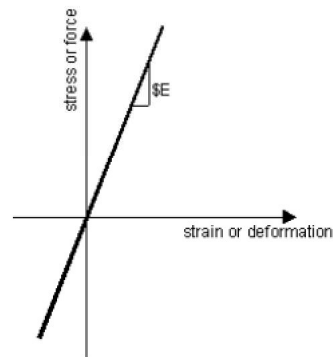
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

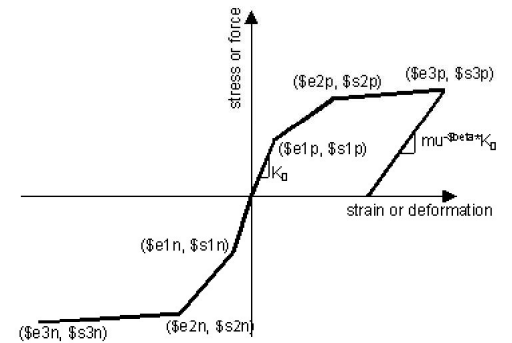
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

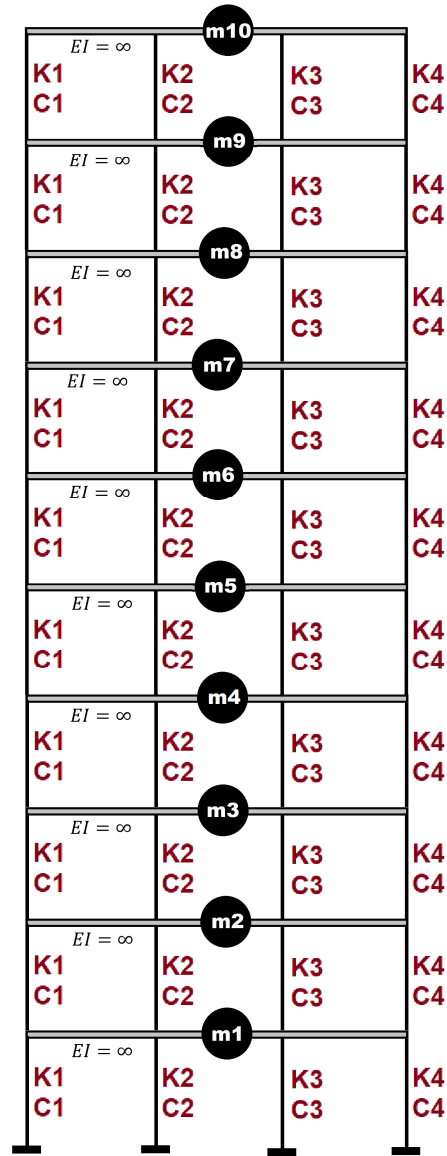
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

Δ_u = Lateral Ultimate Displacement from Pushover Analysis

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...DR\MDOF_10_FLOORS_04_COLUMNS_SEISMIC_EDCI_EVDR.py

MDOF_10_FLOORS_04...ISMIC_EDCI_EVDR.py X

1#####

2#>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

3#FRAGILITY ANALYSIS BASED ON ACCELERATION, STRUCTURAL DUCTILITY DAMAGE INDEX, ENERGY DISSIPATION

4#CAPACITY INDEX, EQUIVALENT VISCOUS DAMPING RATIO WITH NONLINEAR DYNAMIC ANALYSIS

5#OF A MULTI-DEGREE-OF-FREEDOM (MDOF) SYSTEM UTILIZING 200 GROUND MOTIONS IN OPENSEES

6#-----

7#INELASTIC SEISMIC ANALYSIS WITH 200 GROUND MOTIONS

8#-----

9#EQUIVALENT VISCOUS DAMPING RATIO: $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$

10#-----

11#ENERGY DISSIPATION CAPACITY INDEX = $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacement})$

12#-----

13#This program performs Dynamic Analysis on a Multi-Degree-of-Freedom (MDOF) system

14#subjected to 200 seismic ground motions. The analysis evaluates the structural response under varying

15#levels of seismic intensity.

16#The framework is designed to support researchers and engineers in assessing the probabilistic seismic

17#performance of structures, with a focus on understanding the impact of uncertainty on structural

18#response and design.

19#-----

20#Key Features:

21#- Simulation of SDOF system using OpenSees.

22#- Incremental scaling of ground motions for Nonlinear Dynamic Analysis.

23#- Probabilistic fragility assessment based on predefined damage states.

24#- Visualization of structural response and fragility curves.

25#- Export of results for further analysis.

26#-----

27#THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

28#EMAIL: salar.d.ghashghaei@gmail.com

29#####

30

31

32This code implements a comprehensive nonlinear dynamic analysis framework for

33performance-based earthquake engineering assessment of single-degree-of-freedom

34(SDOF) systems. The methodology combines traditional nonlinear time-history

...OLUMNS_8_ANA\MDOF_10_FLOORS_04_COLUMNS_SEISMIC_EDCI_EVDR

32 %

3D CONTOUR: Max. Displacement [m] - Max. Velocity [m/s] - Max. Acceleration [m/s^2]

IPython Console

Plots

Variable Explorer

Help

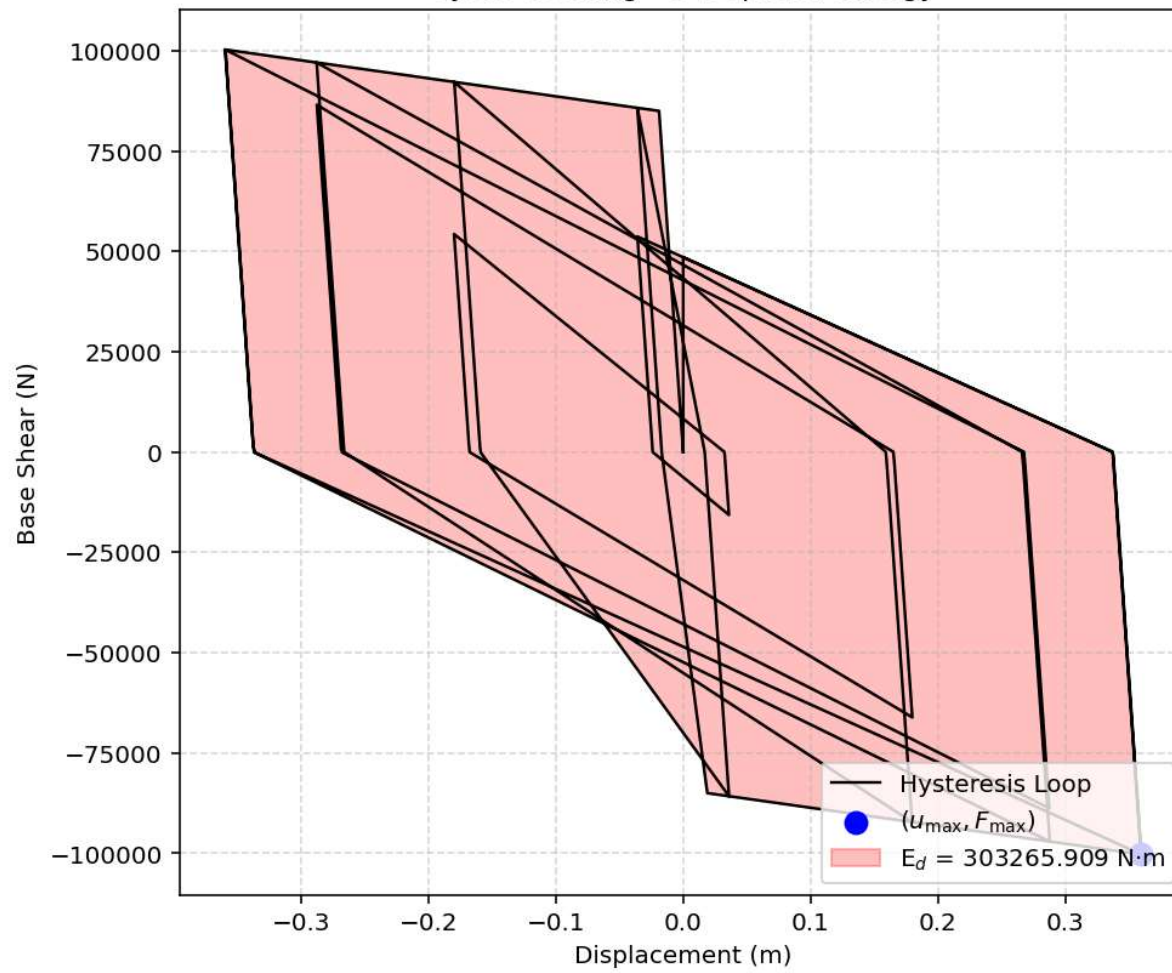
Debugger

History

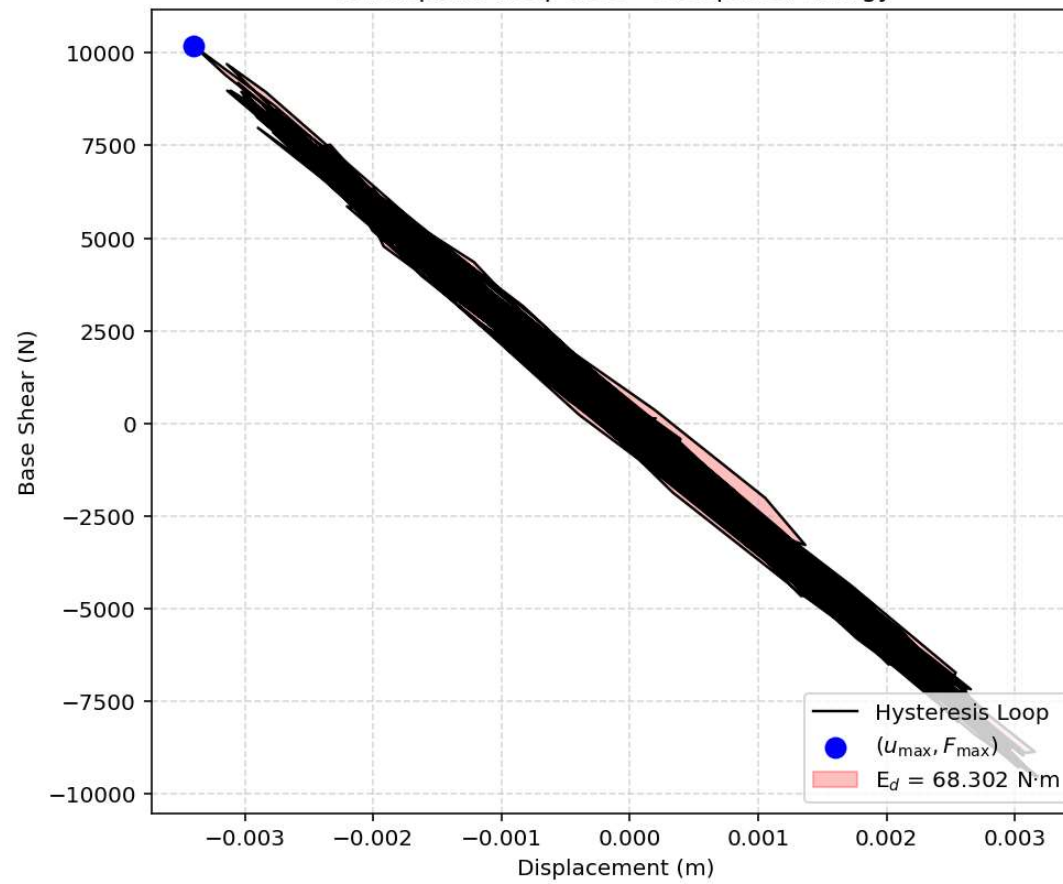
Files

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1224, Col 22 UTF-8 CRLF RW Mem 49%

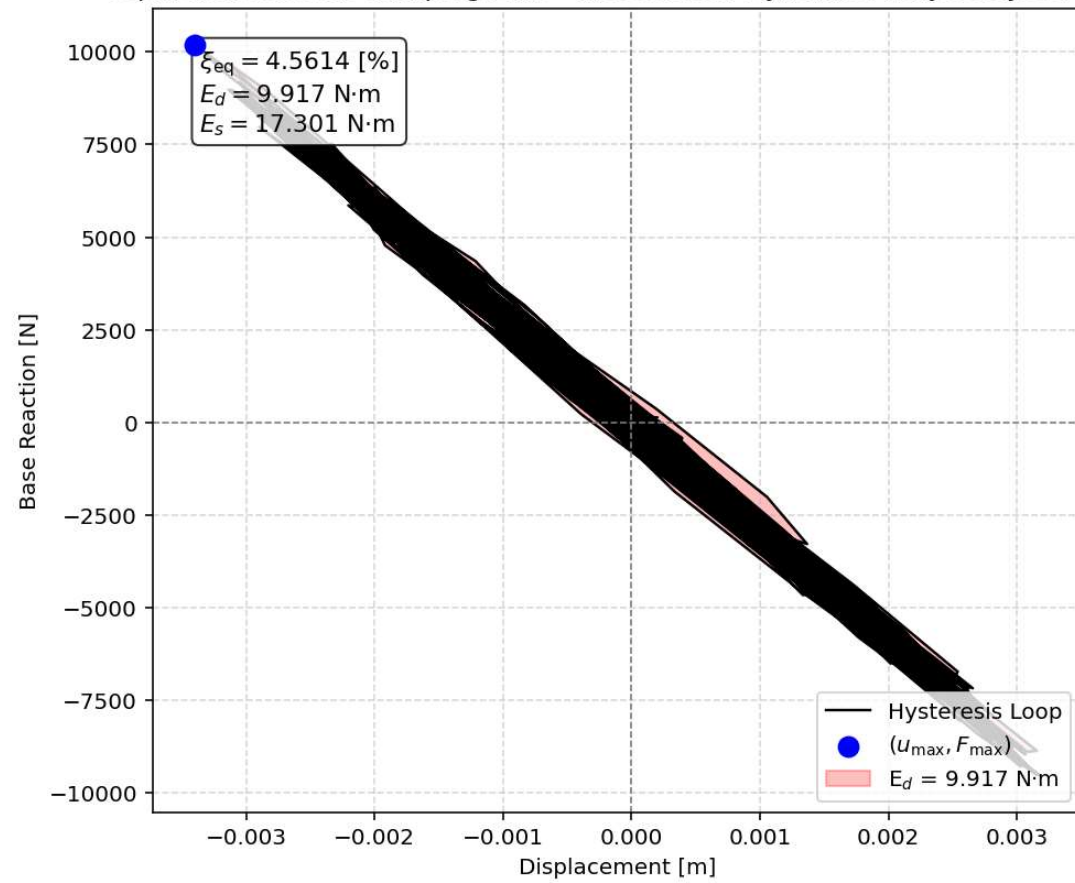
Cyclic Loading - Dissipated Energy



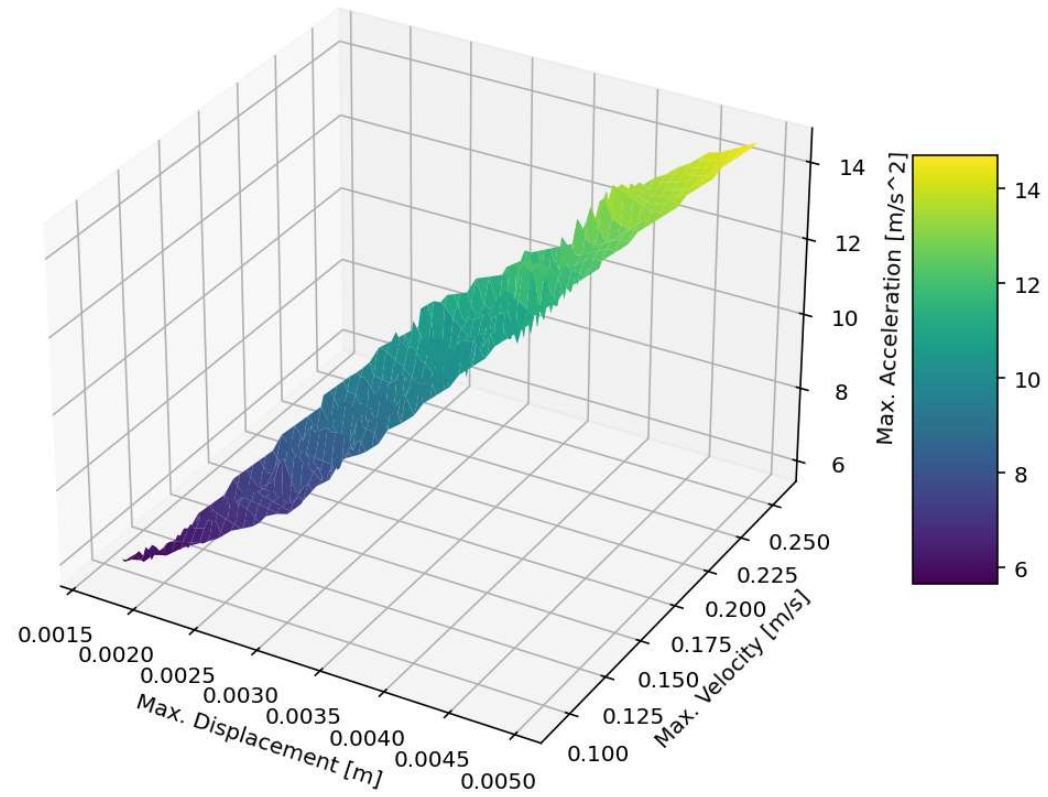
Earthquake Response - Dissipated Energy

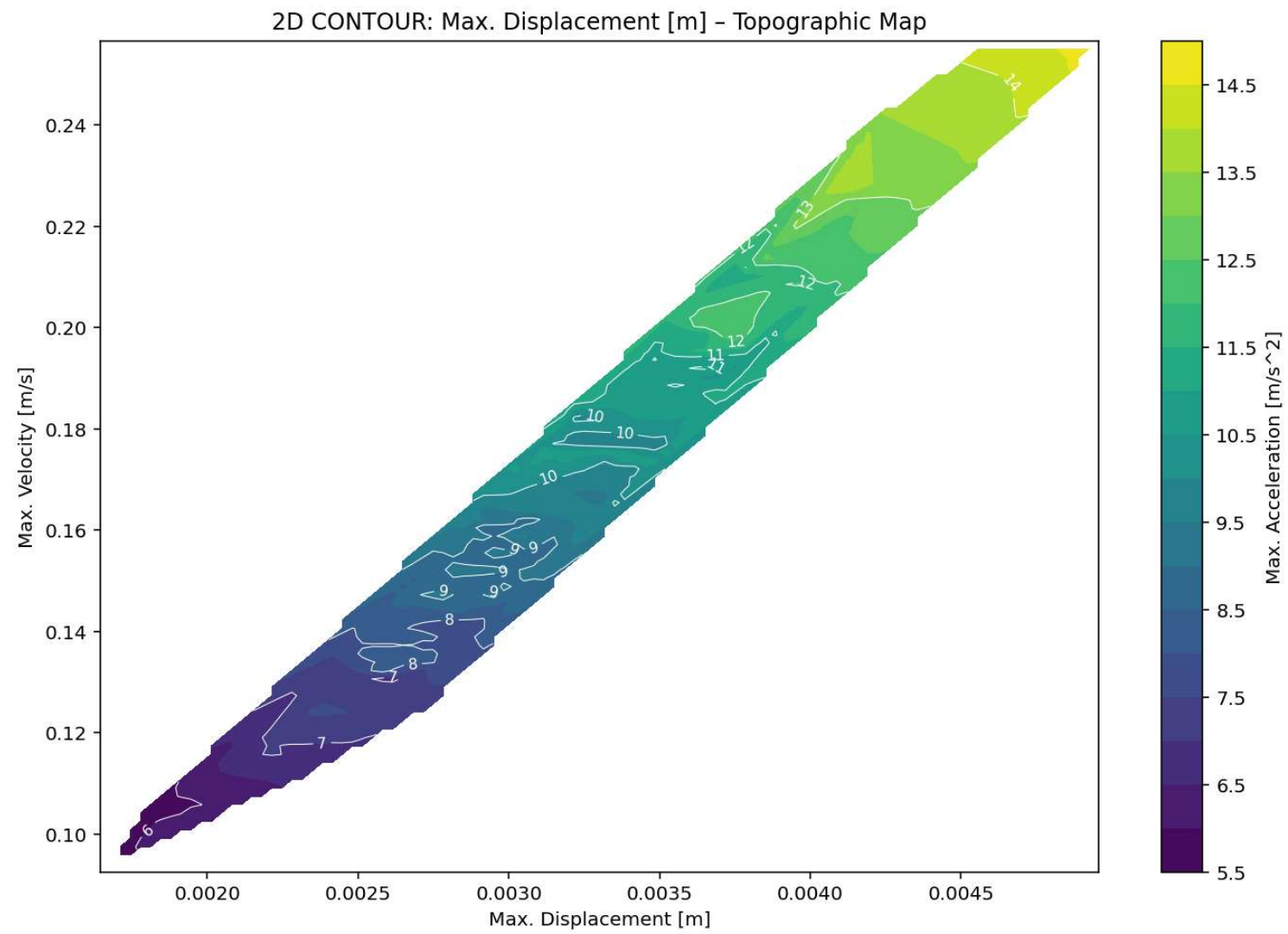


Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis

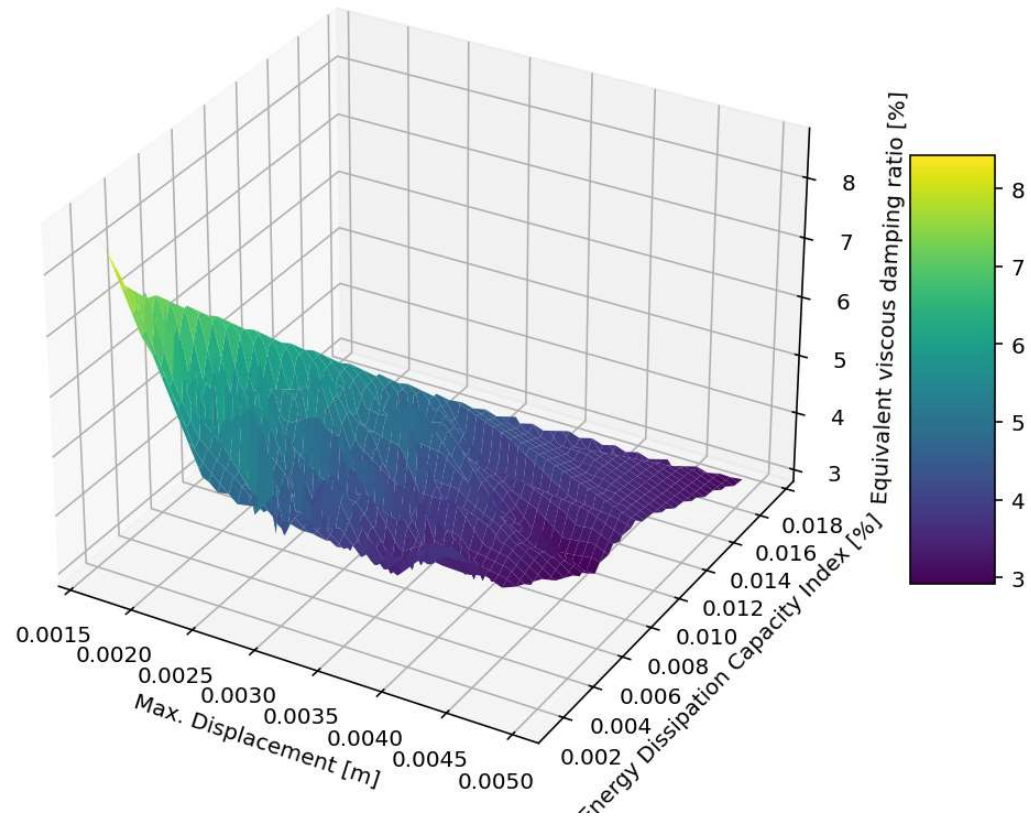


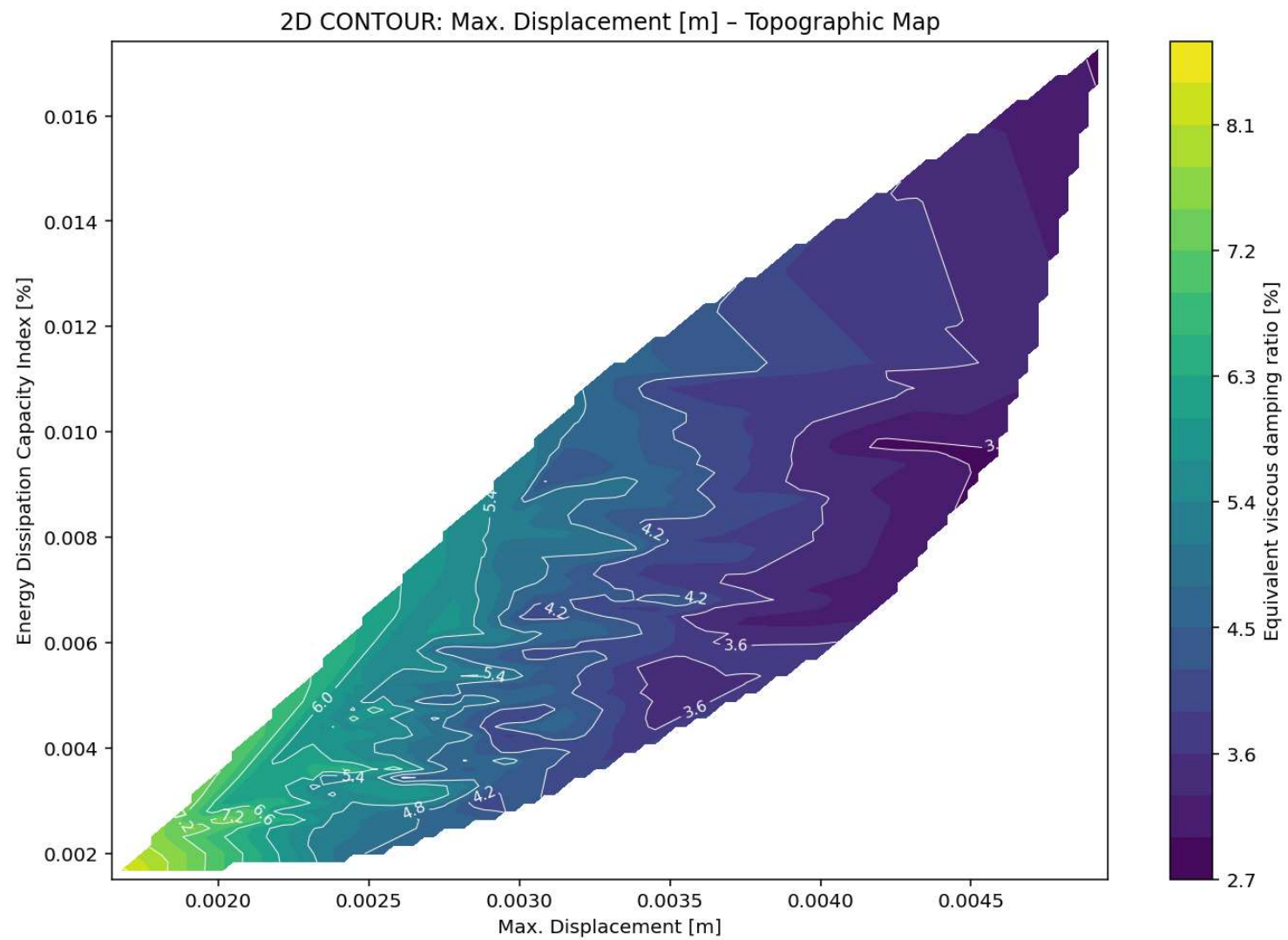
3D CONTOUR: Max. Displacement [m] - Max. Velocity [m/s] - Max. Acceleration [m/s²]



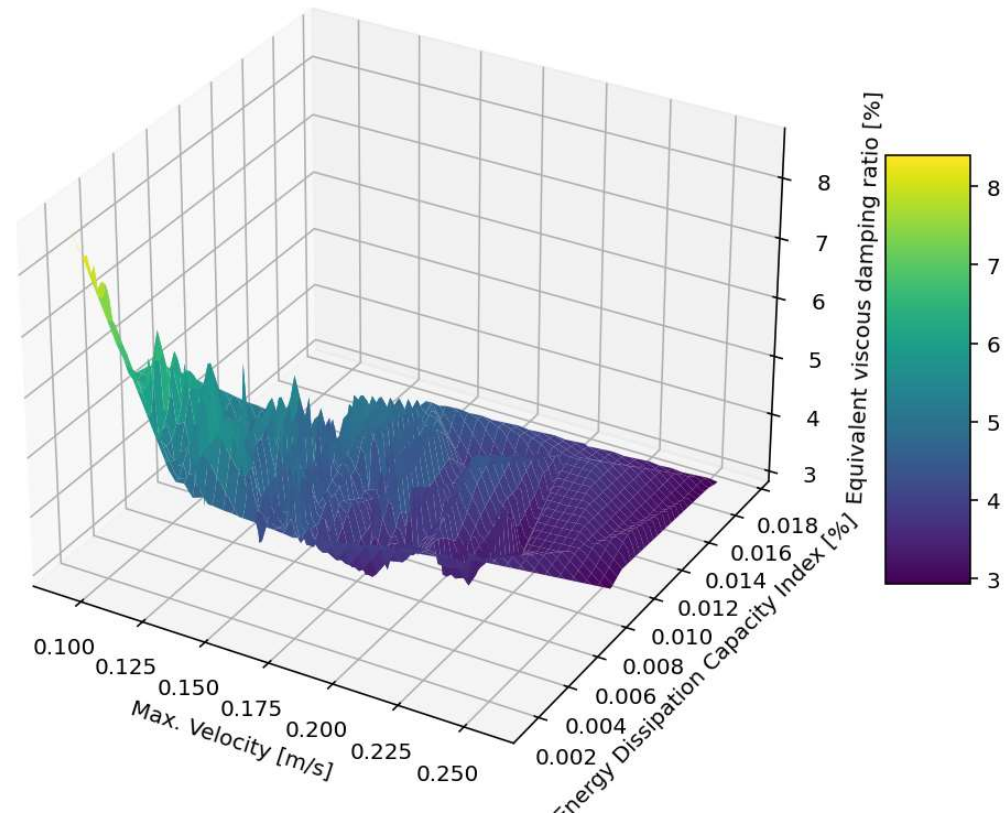


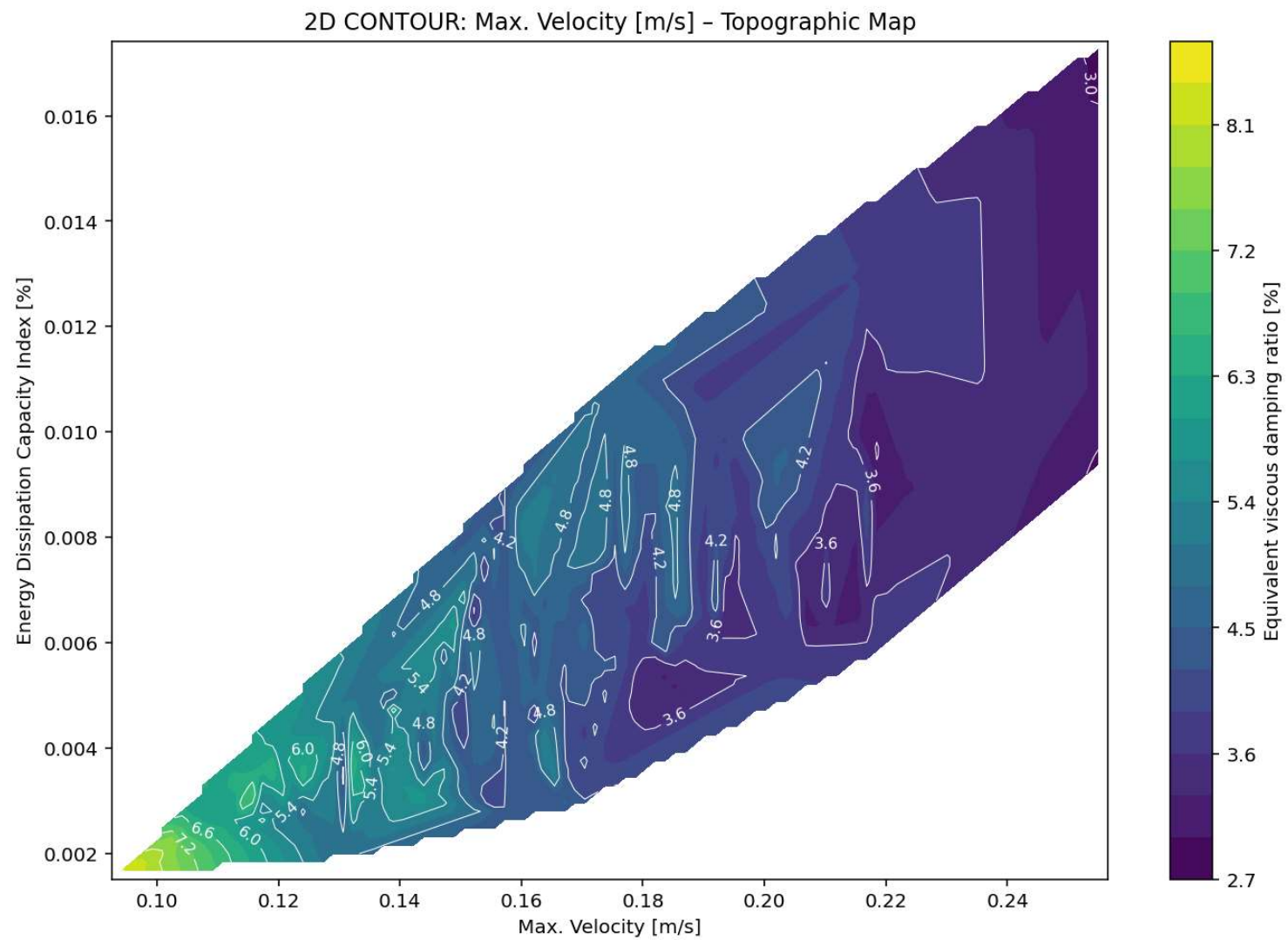
3D CONTOUR: Max. Displacement [m] - Energy Dissipation Capacity Index [%] - Equivalent viscous damping ratio [%]





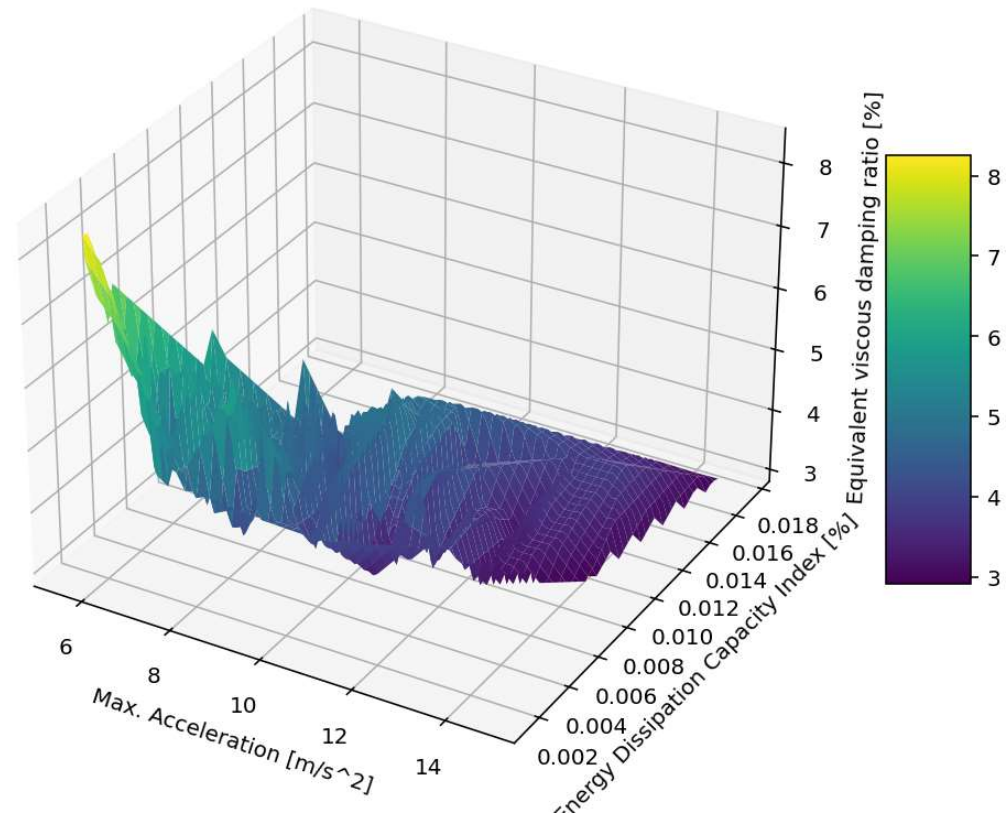
3D CONTOUR: Max. Velocity [m/s] - Energy Dissipation Capacity Index [%] - Equivalent viscous damping ratio [%]



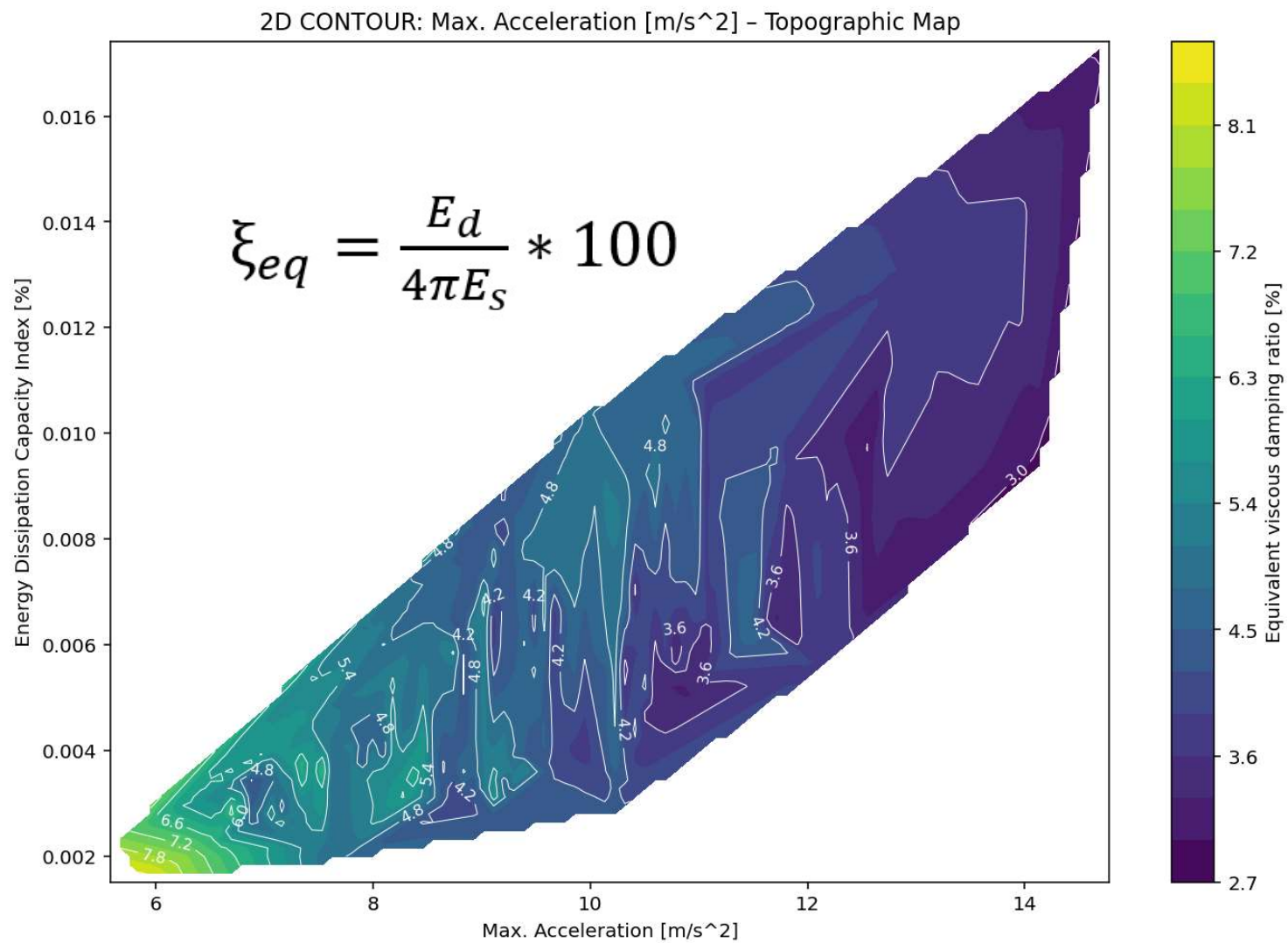


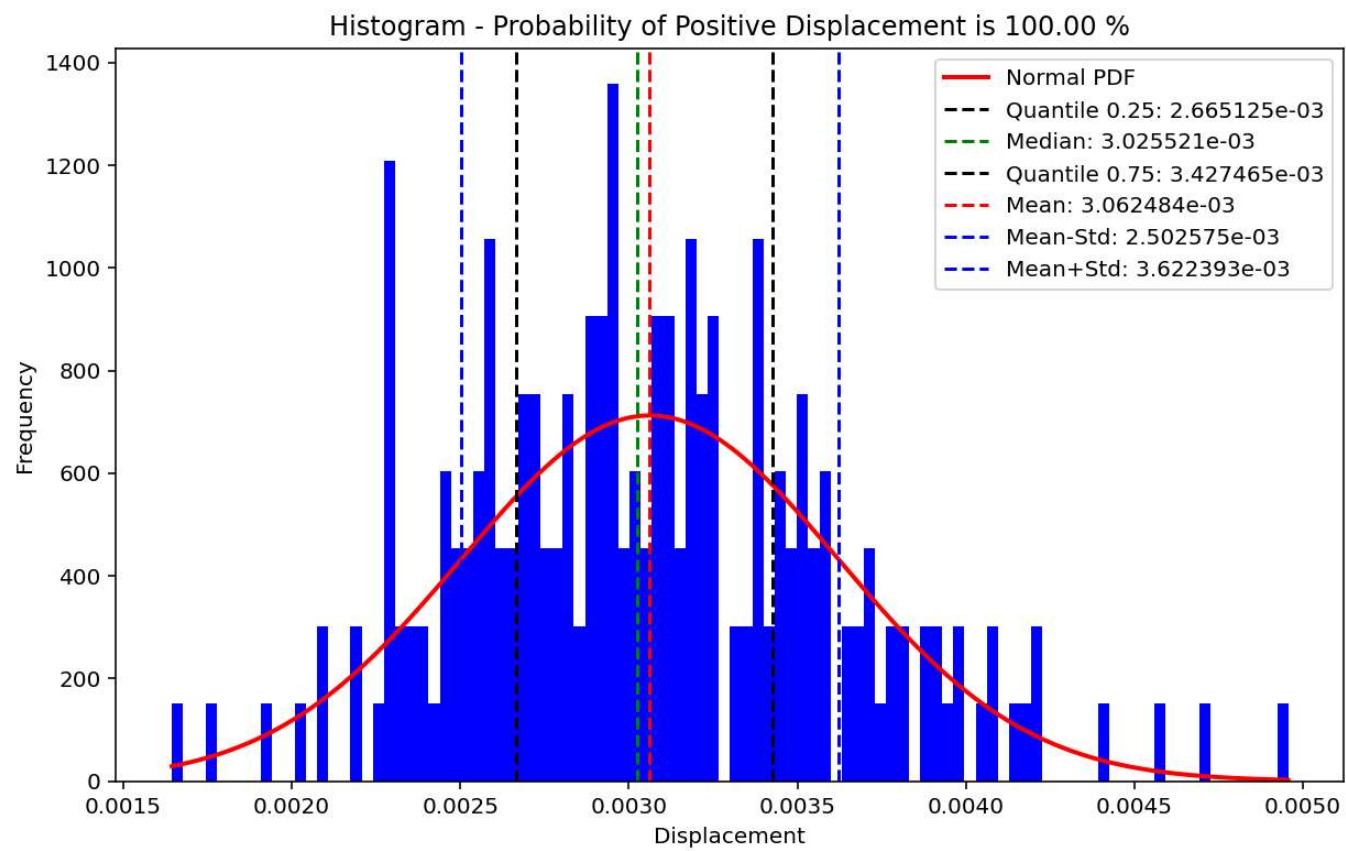
3D CONTOUR: Max. Acceleration [m/s²] - Energy Dissipation Capacity Index [%] - Equivalent viscous damping ratio [%]

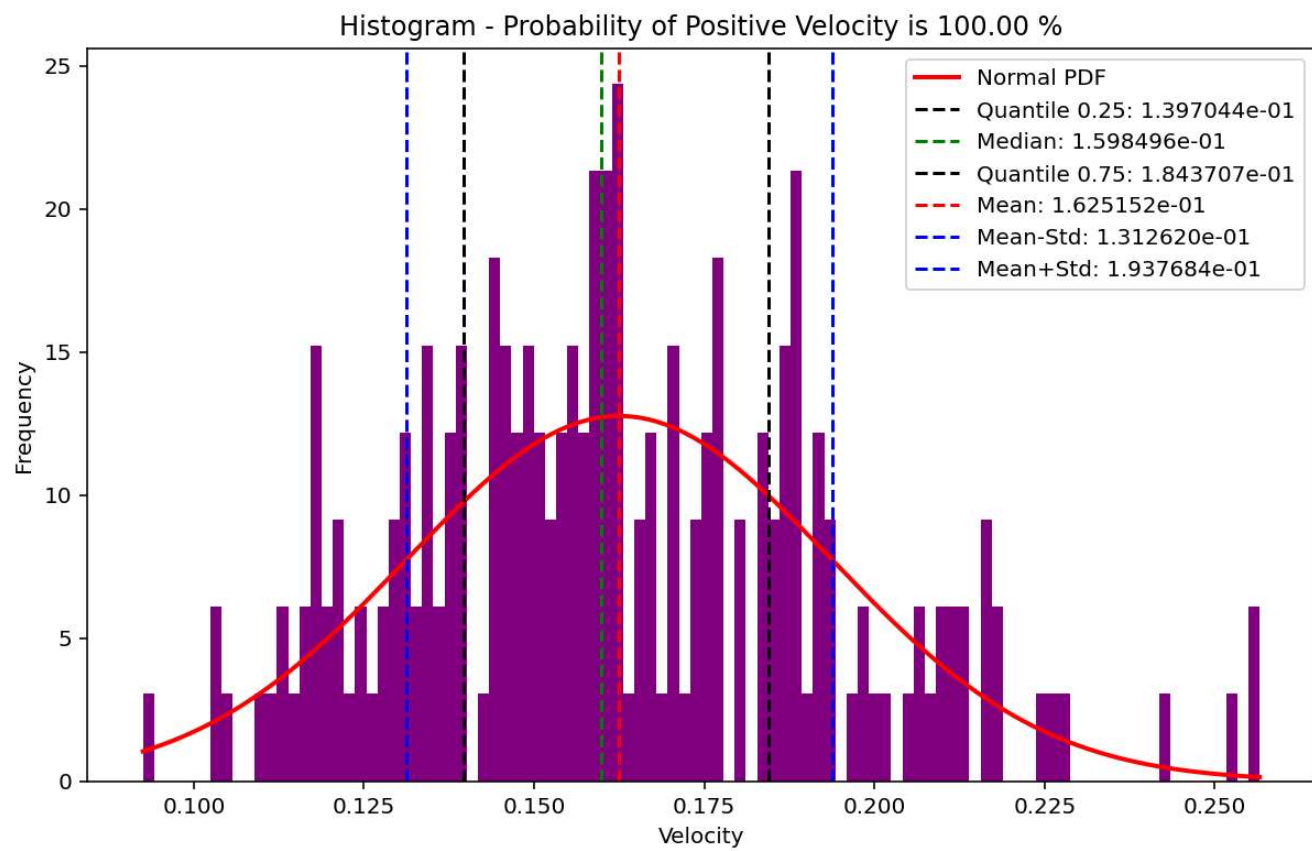
$$\xi_{eq} = \frac{E_d}{4\pi E_s} * 100$$

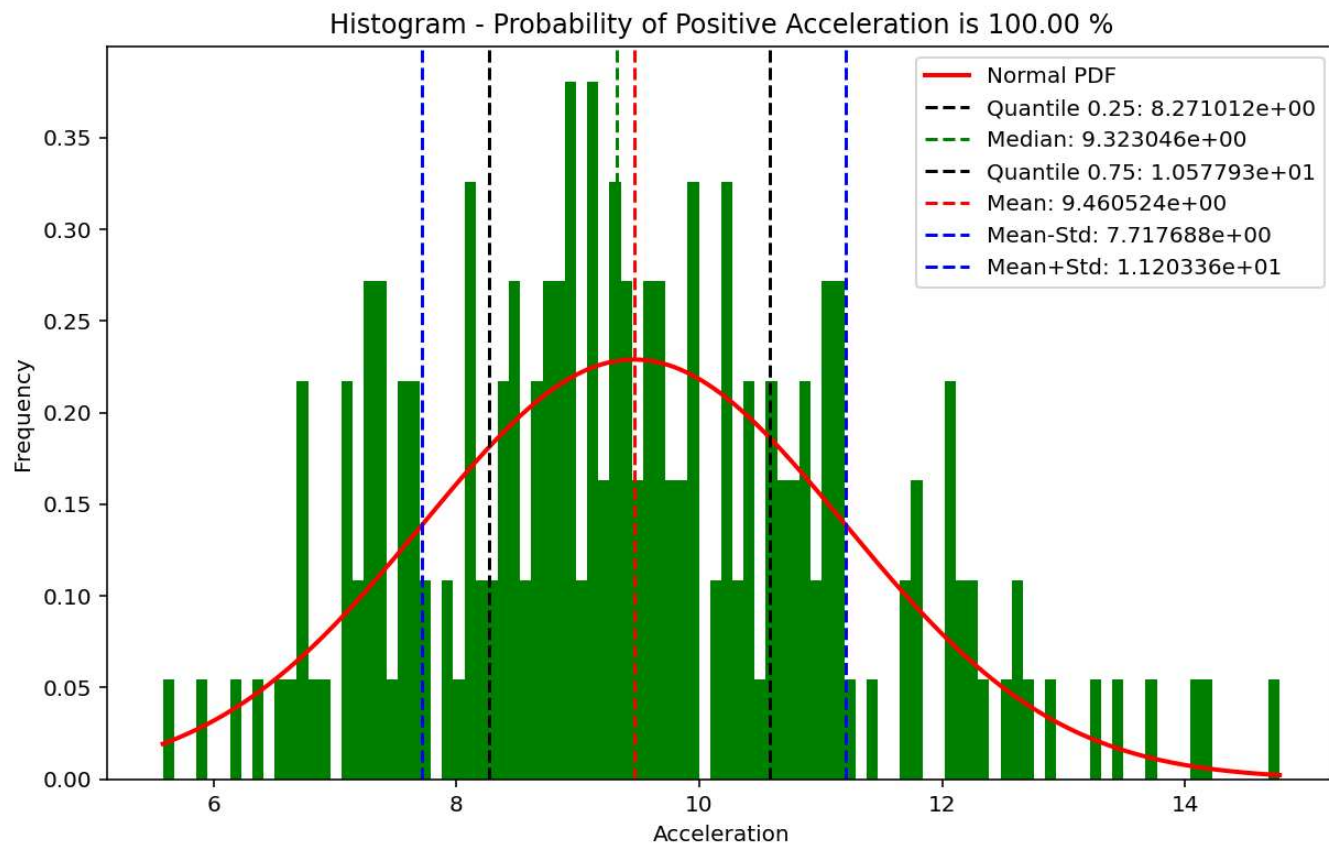


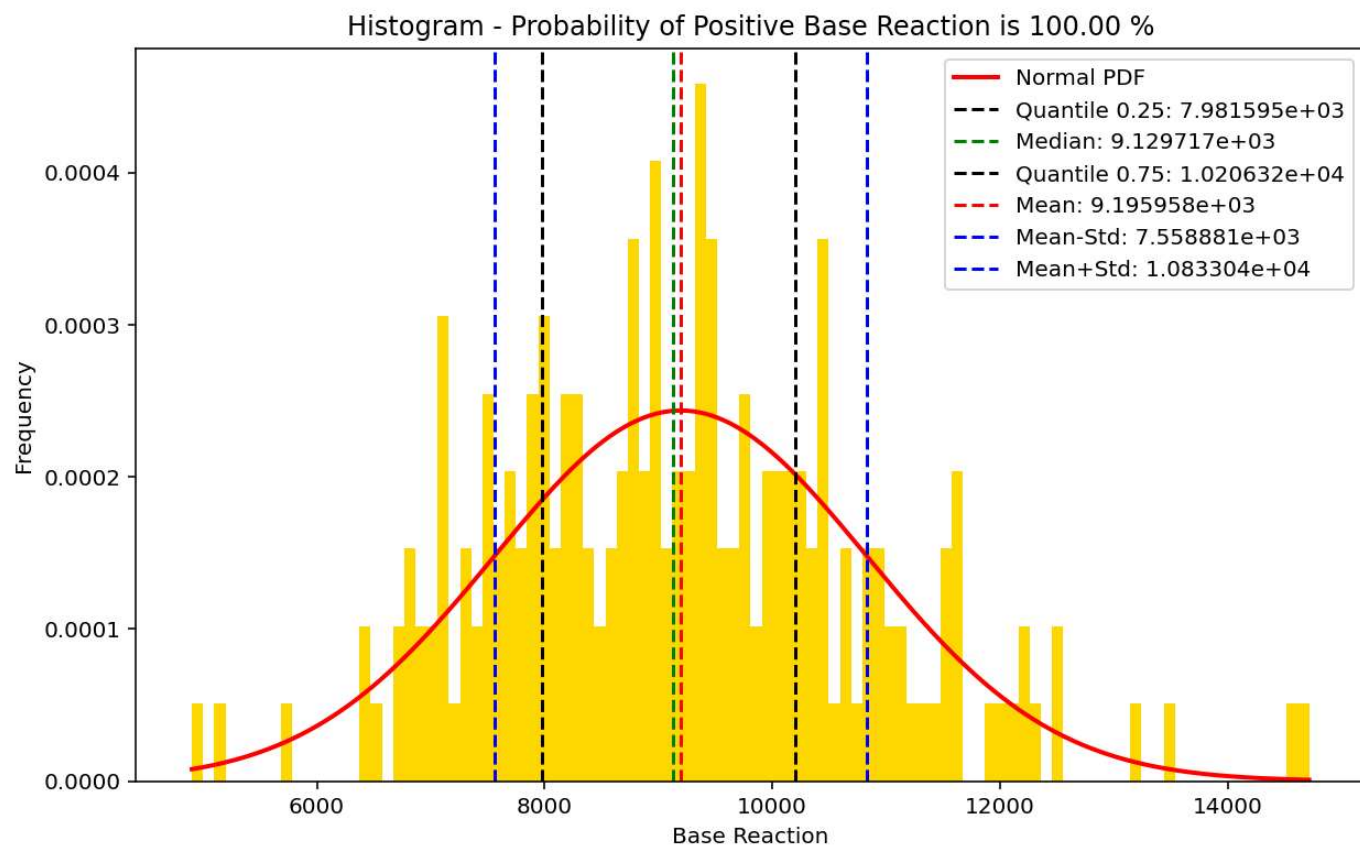
$$\text{Energy Dissipation Capacity Index} = \frac{E_d(\text{Dissipated Energy from Earthquake})}{E_d(\text{Dissipated Energy from Cyclic Displacement})} * 100$$

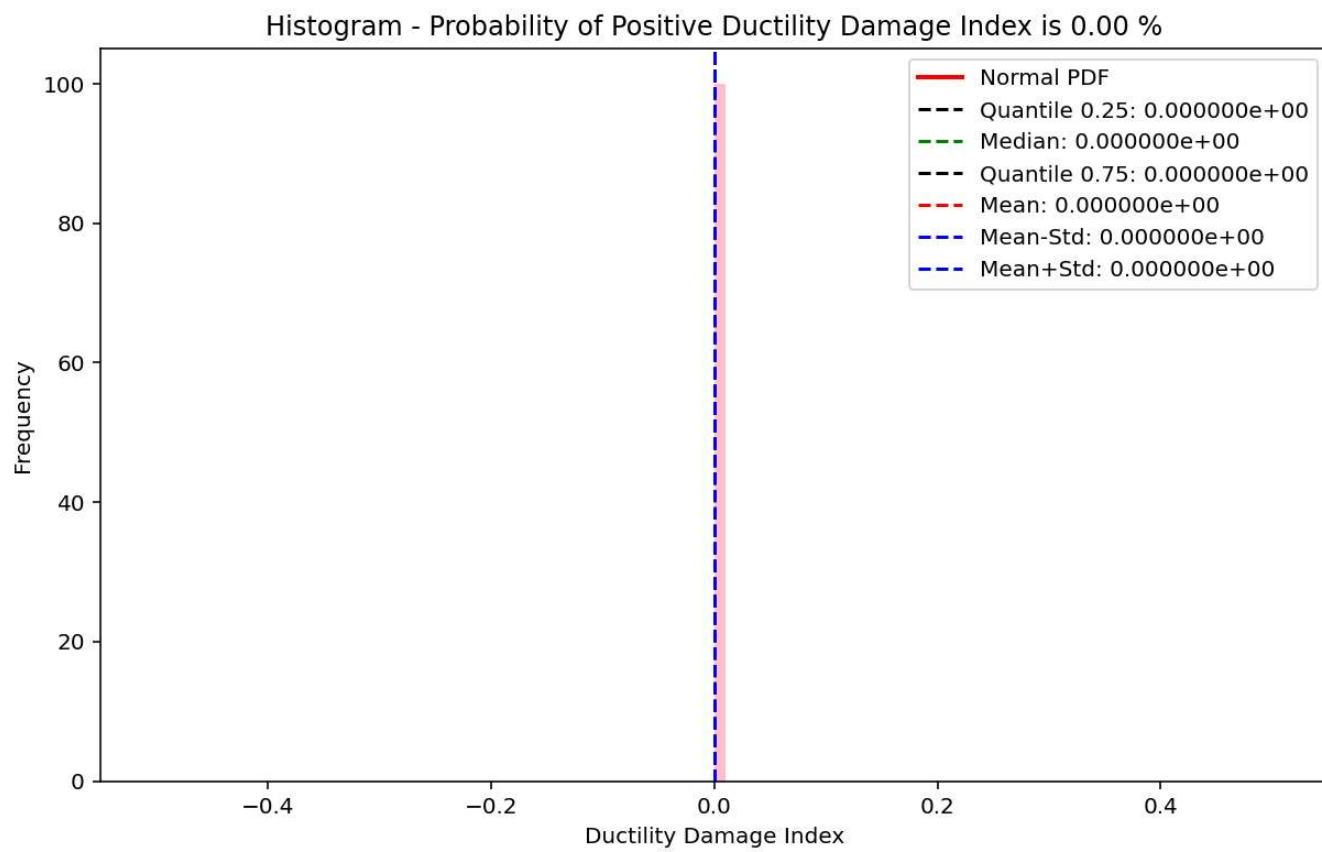


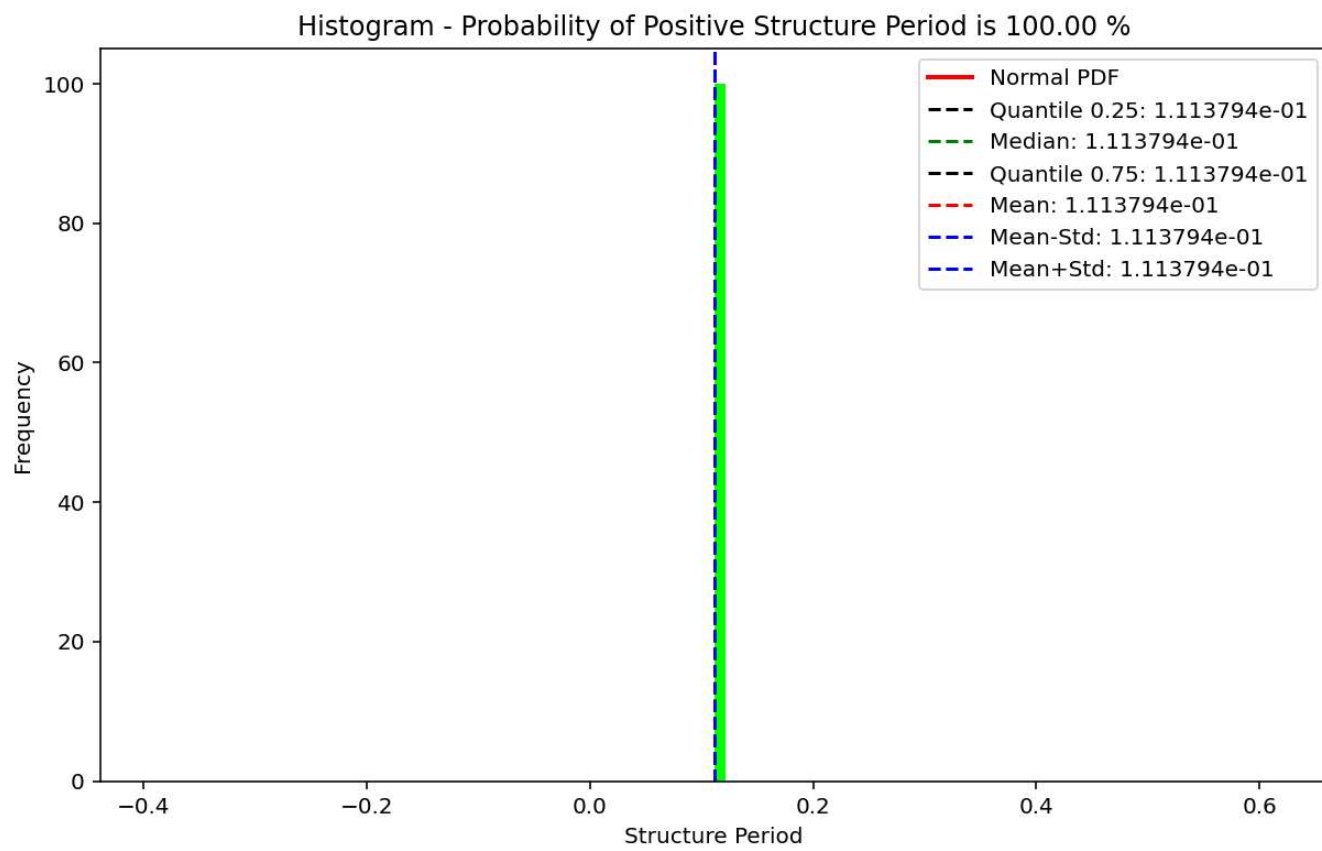


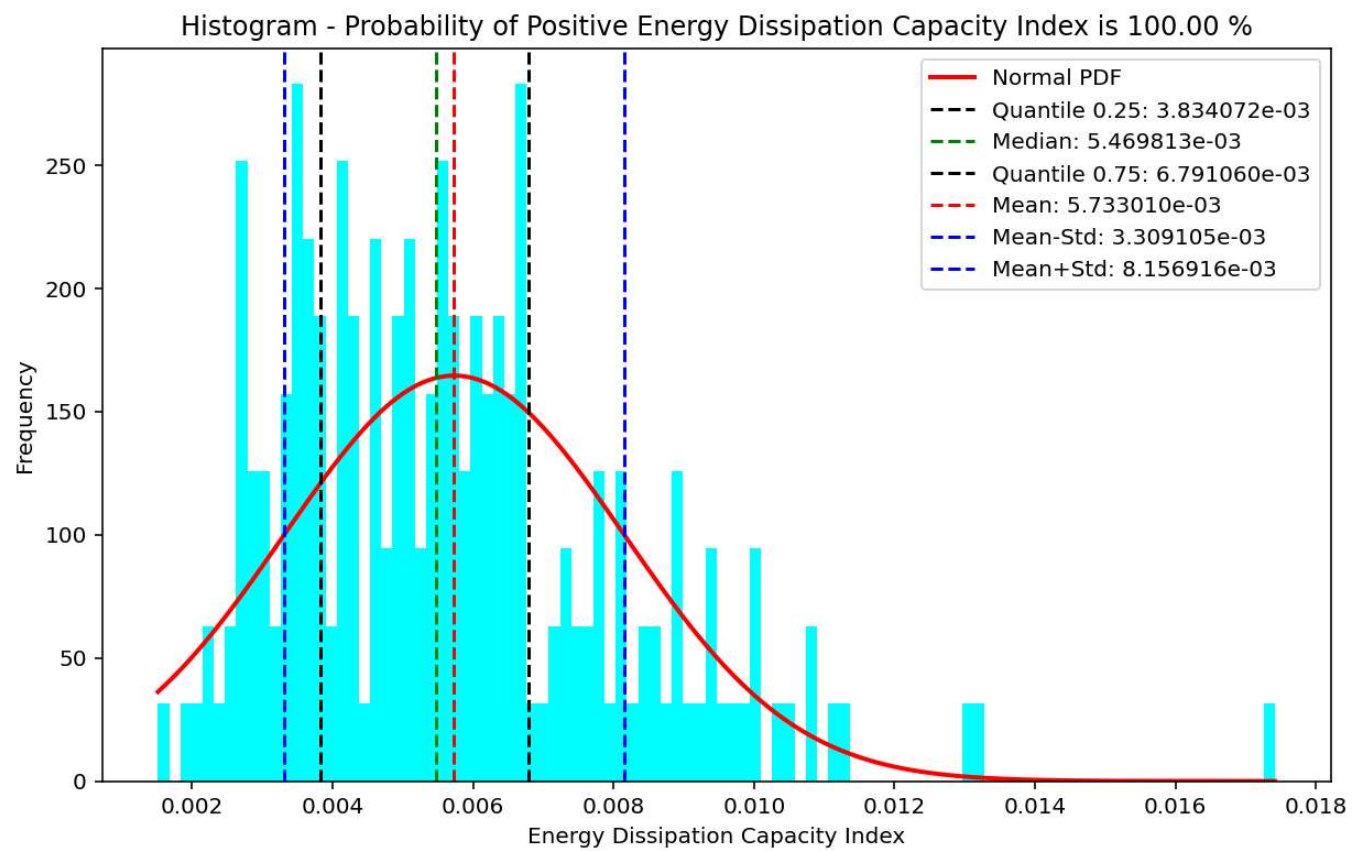


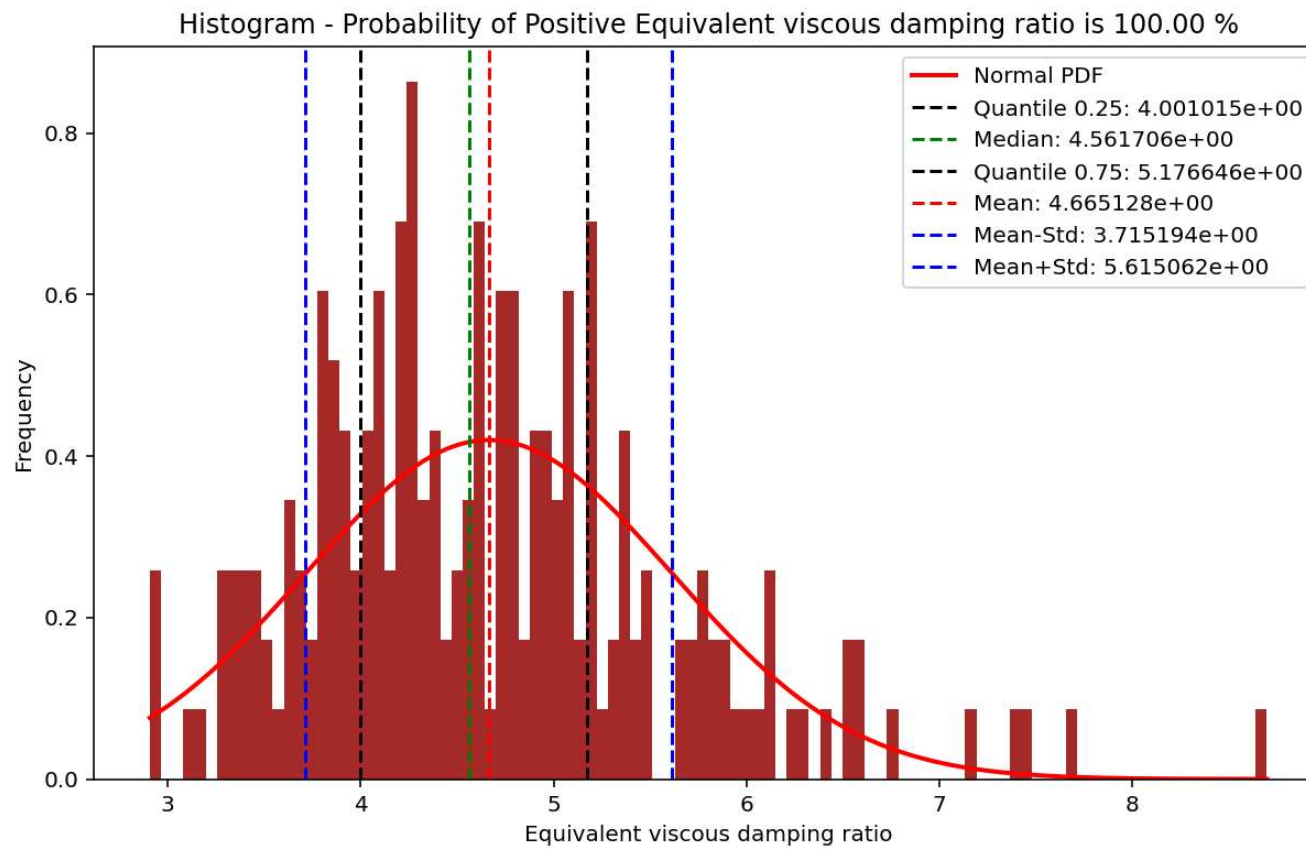


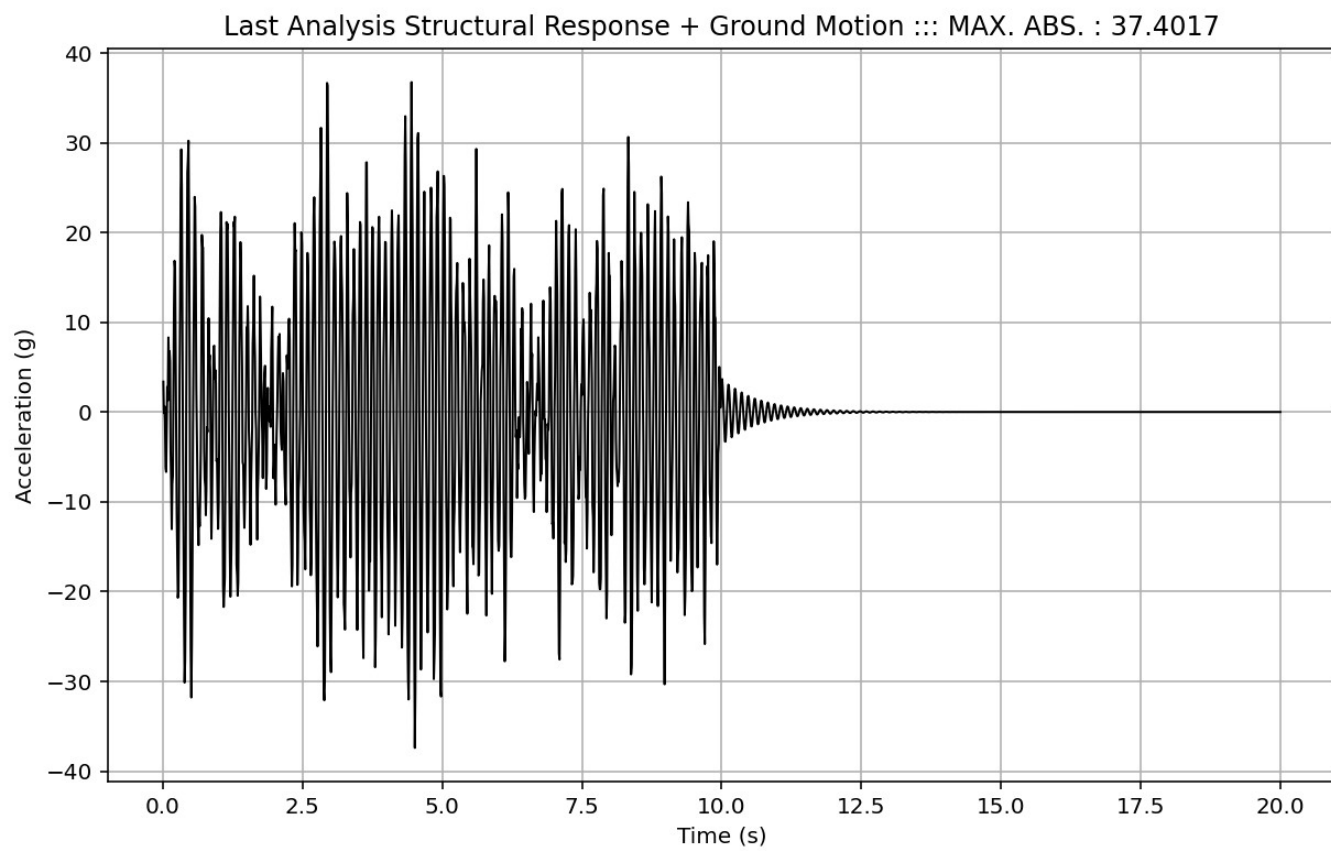


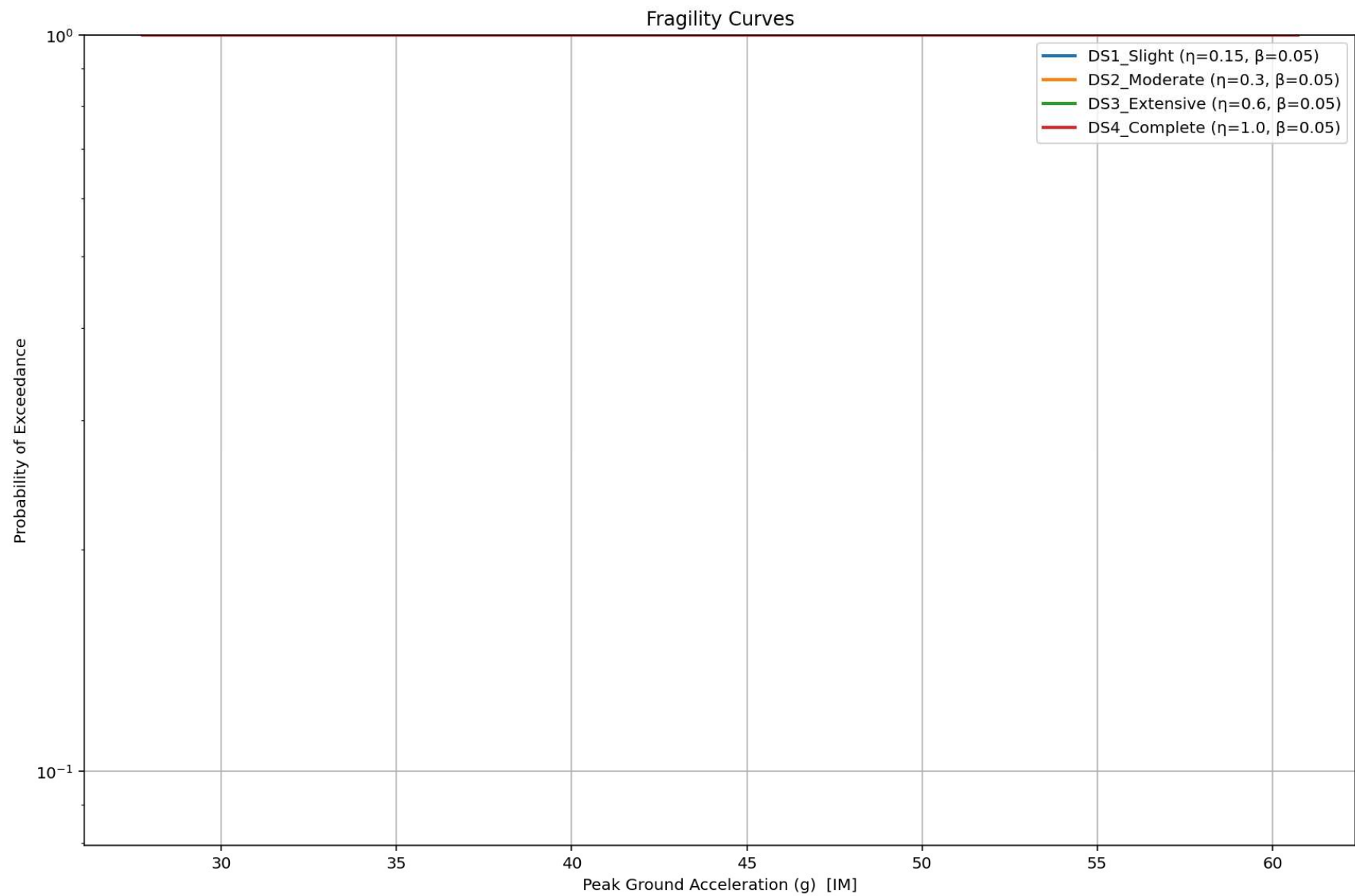




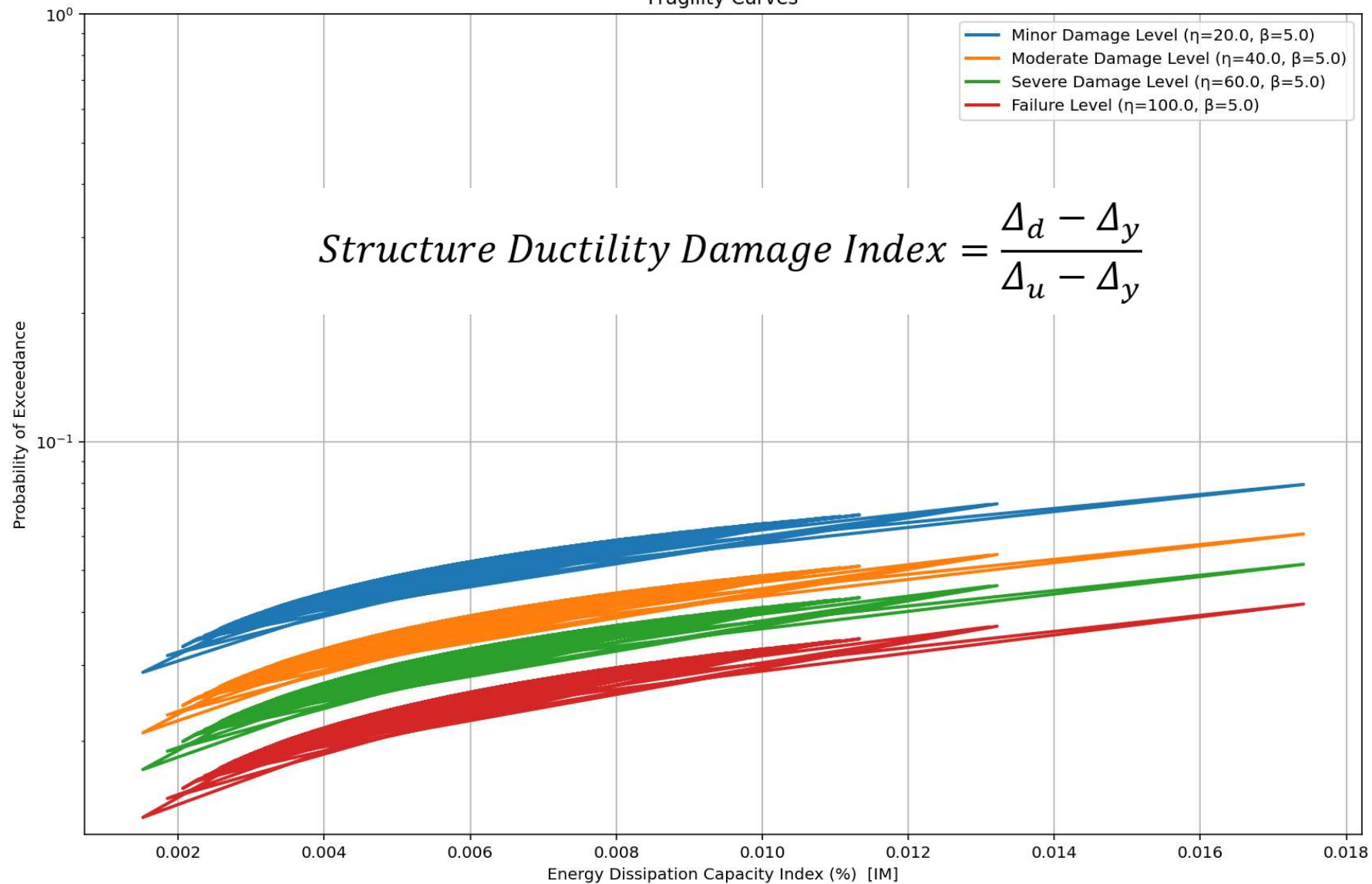


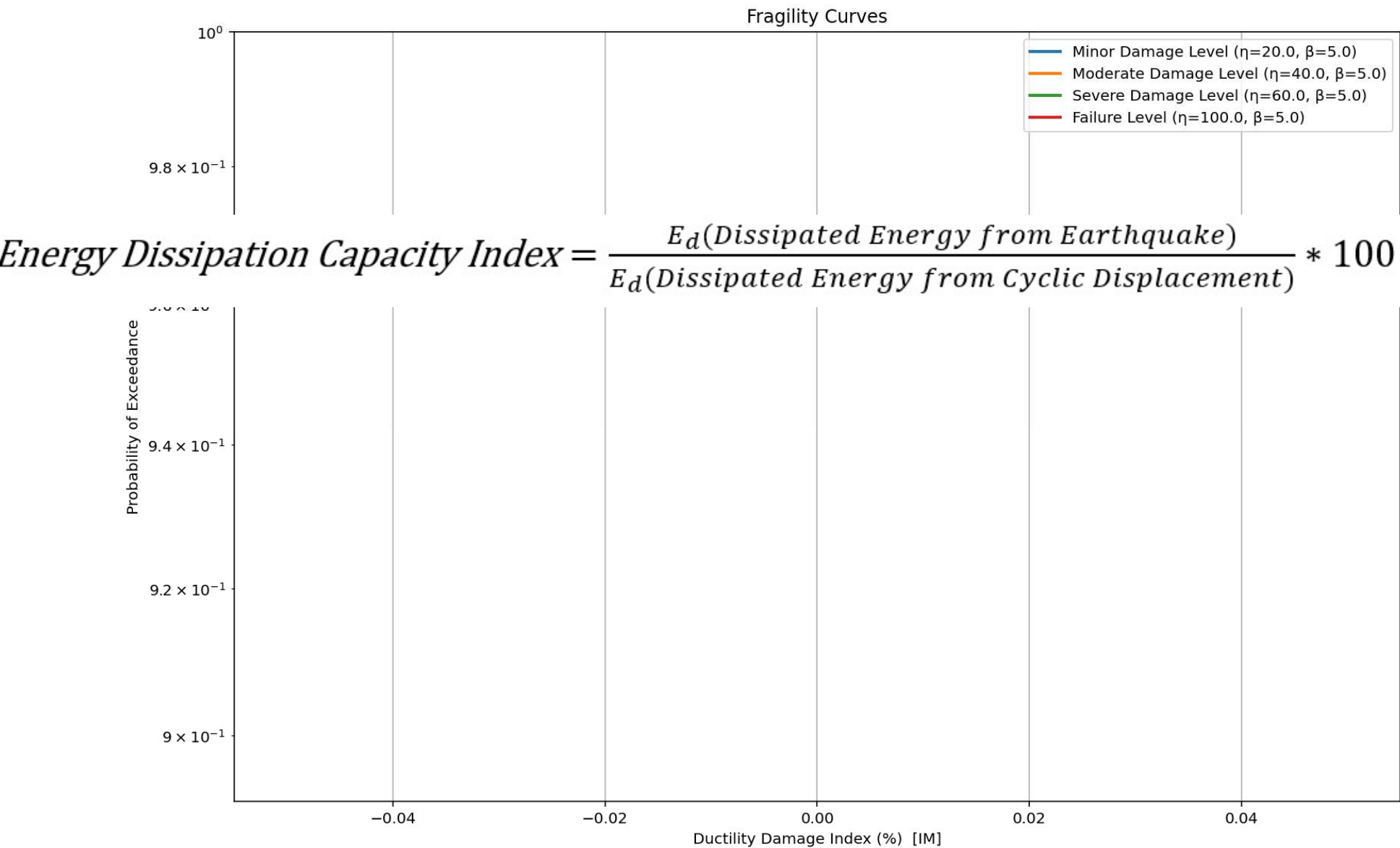




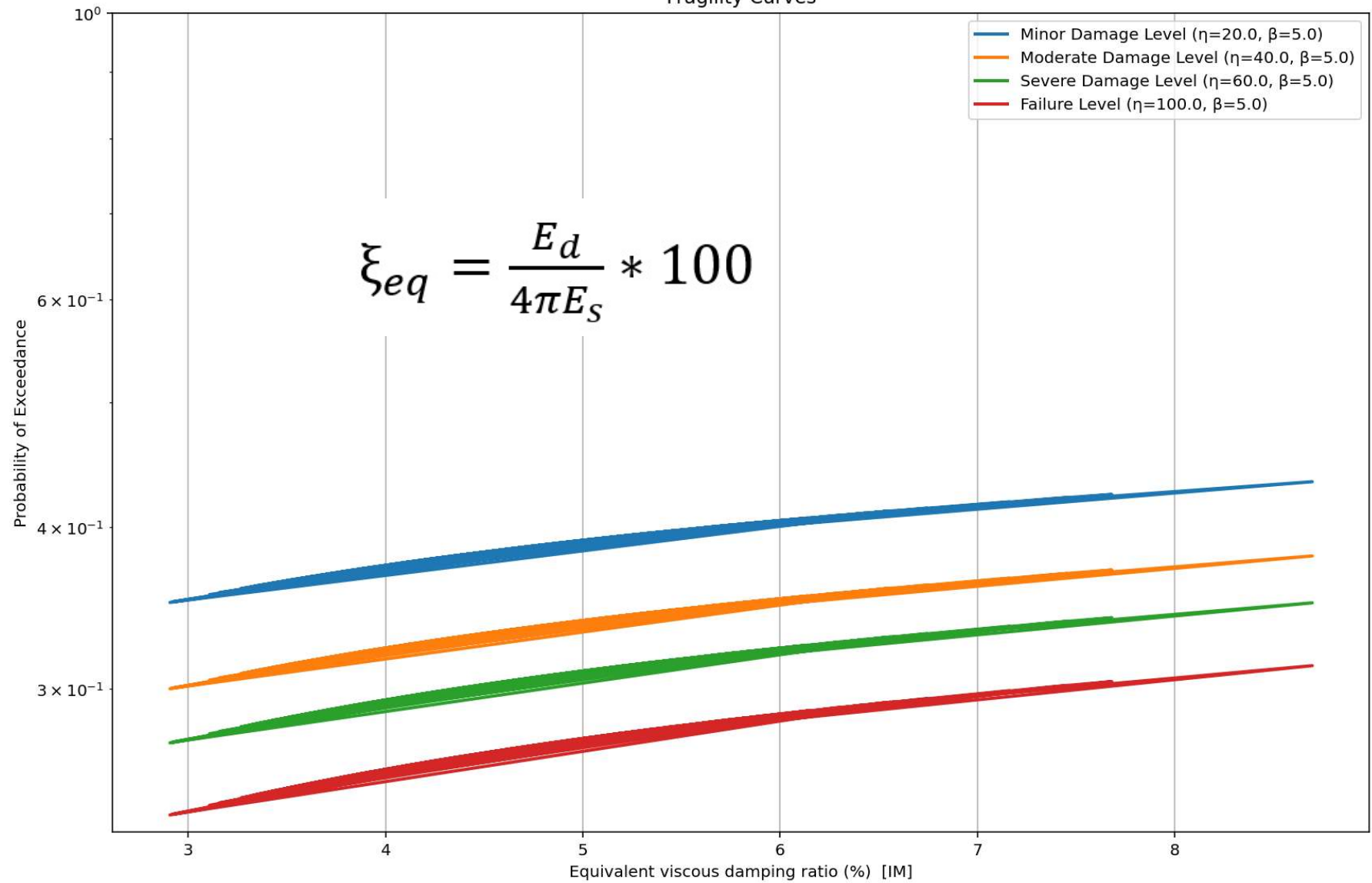


Fragility Curves

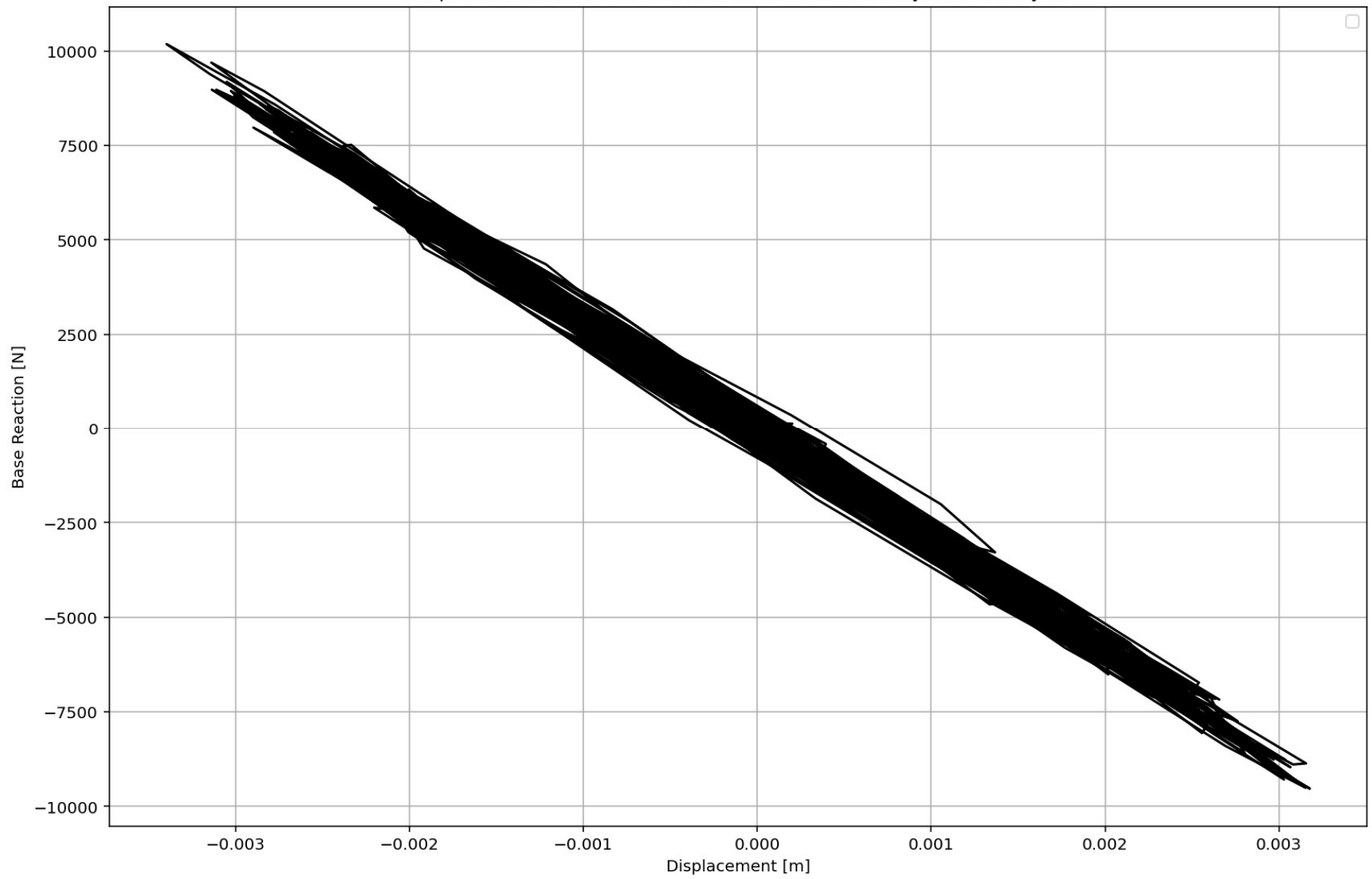


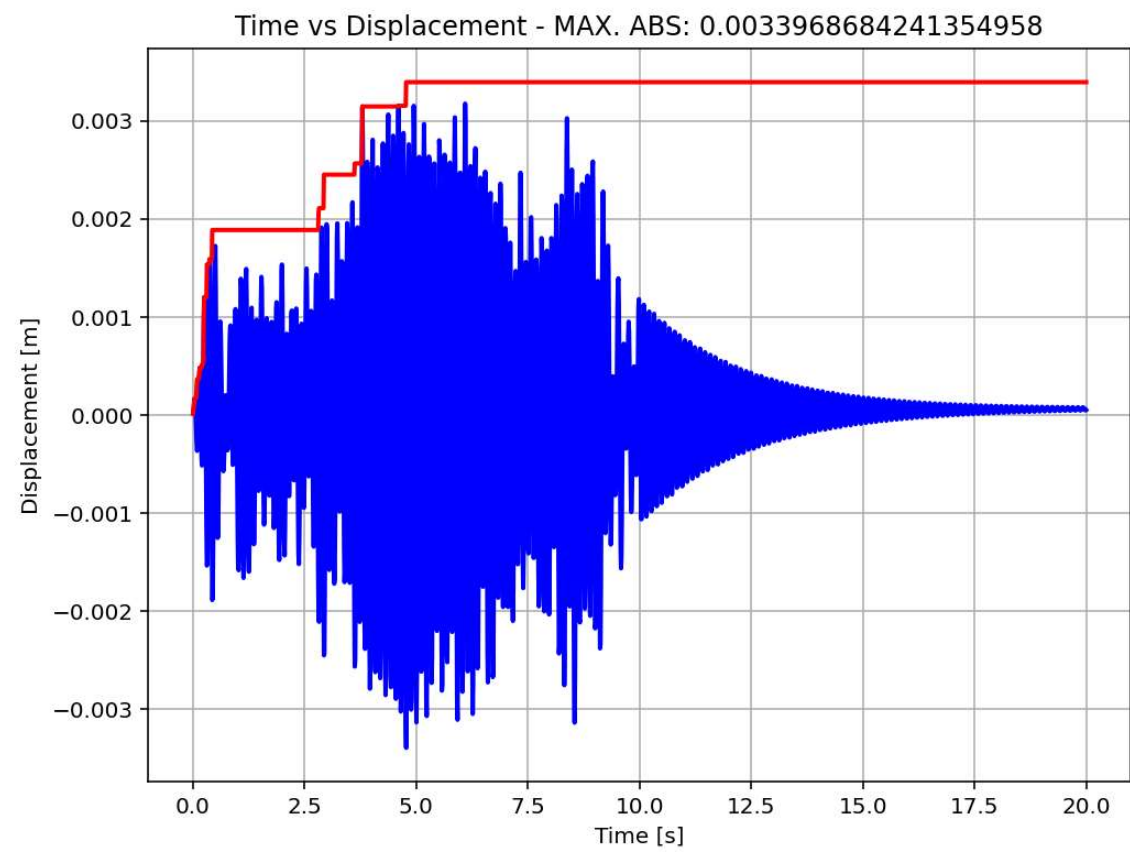


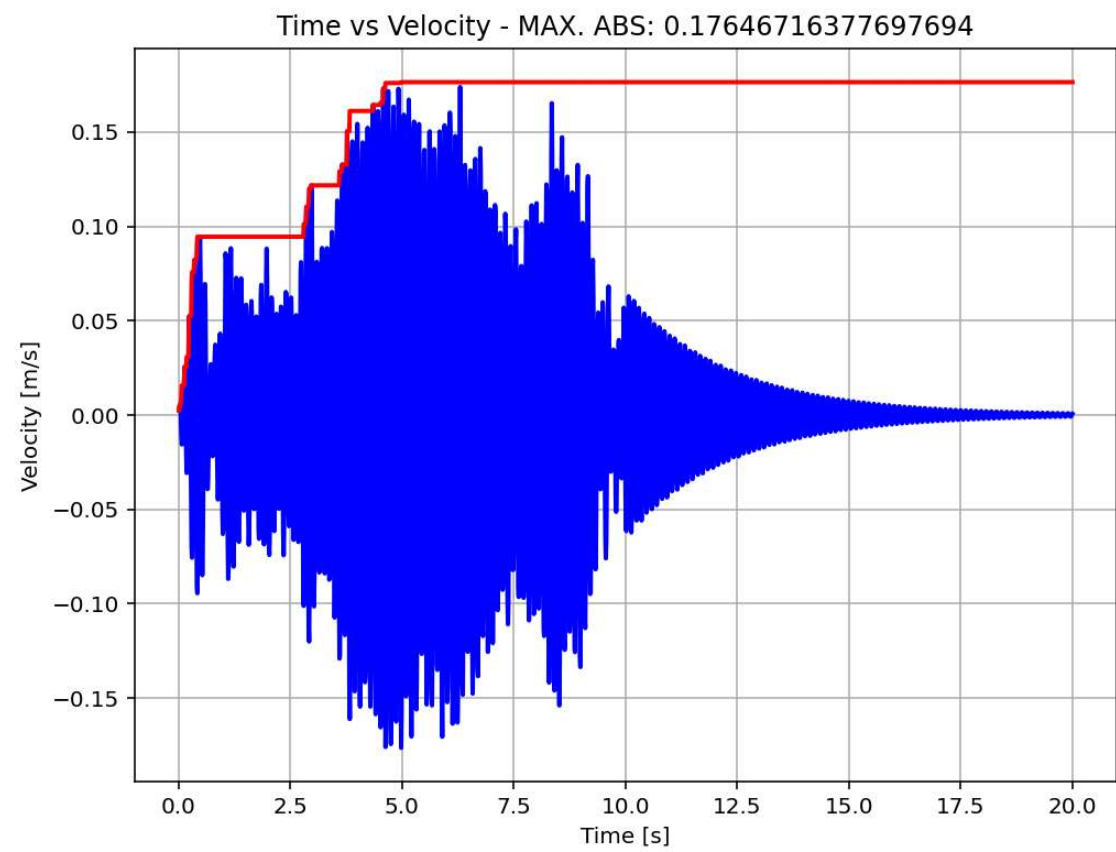
Fragility Curves

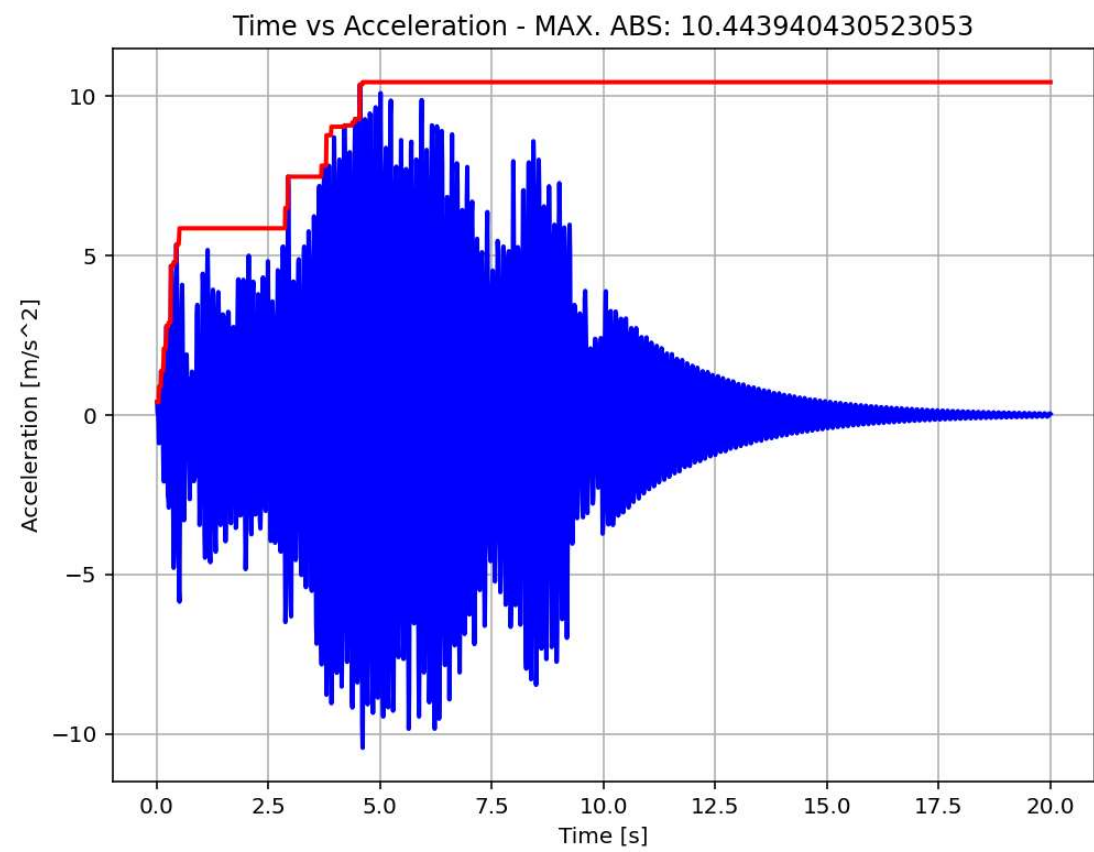


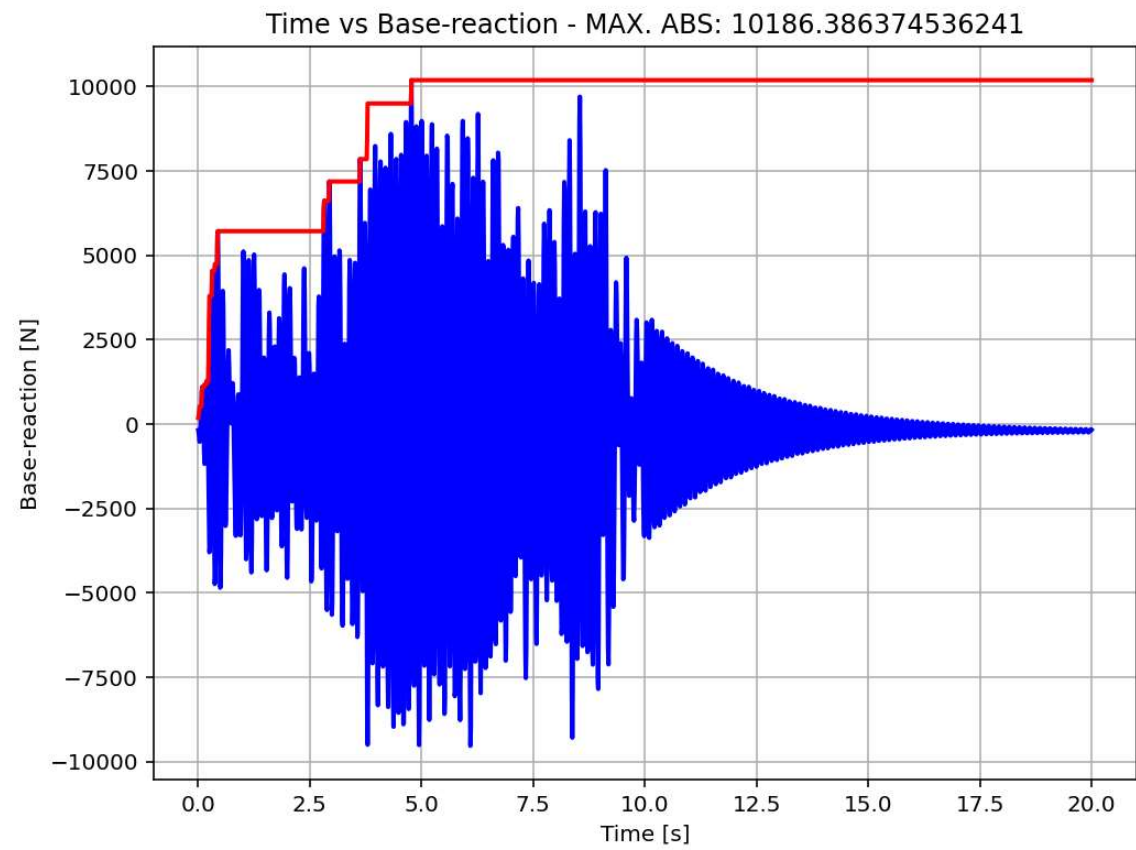
Displacement & Base Reaction Relation From Last Dynamic Analysis



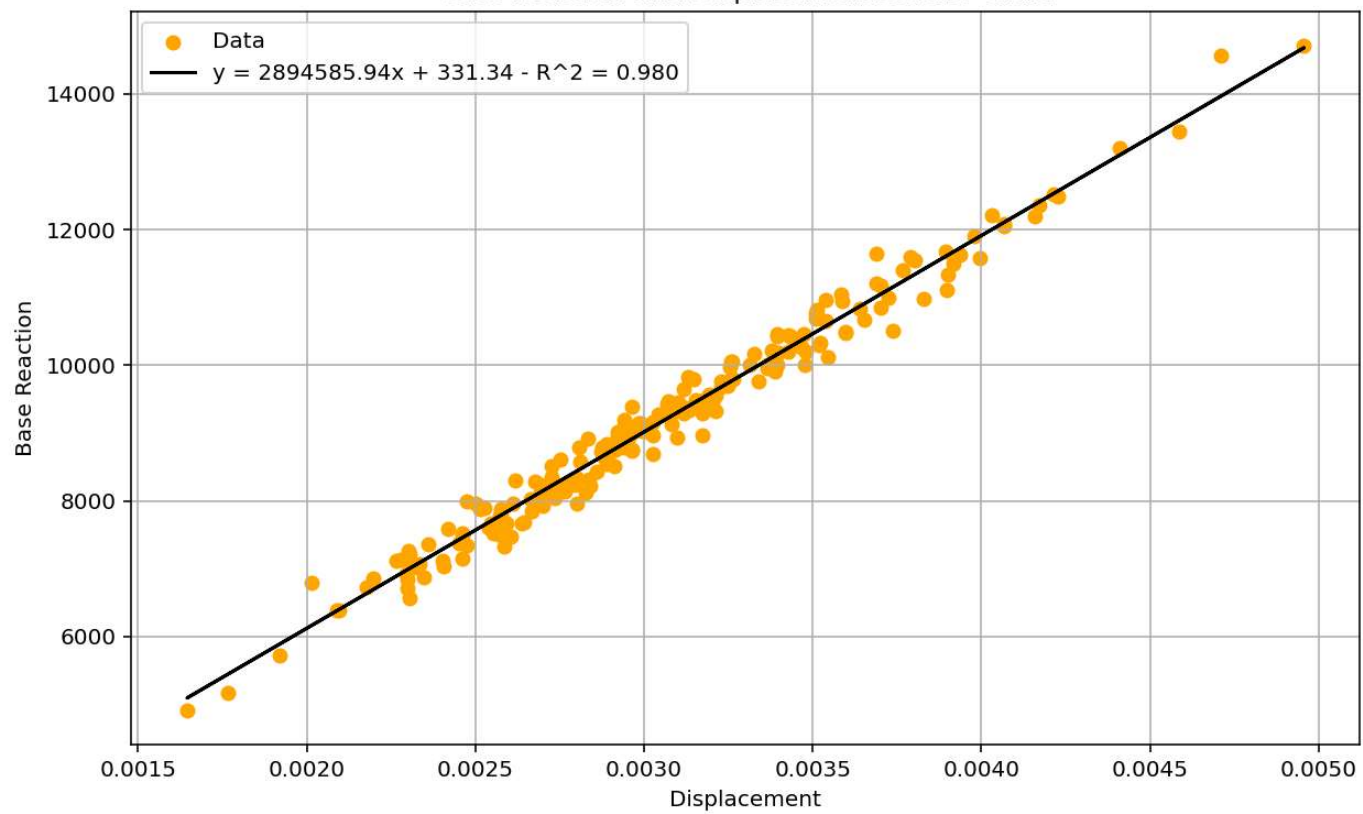


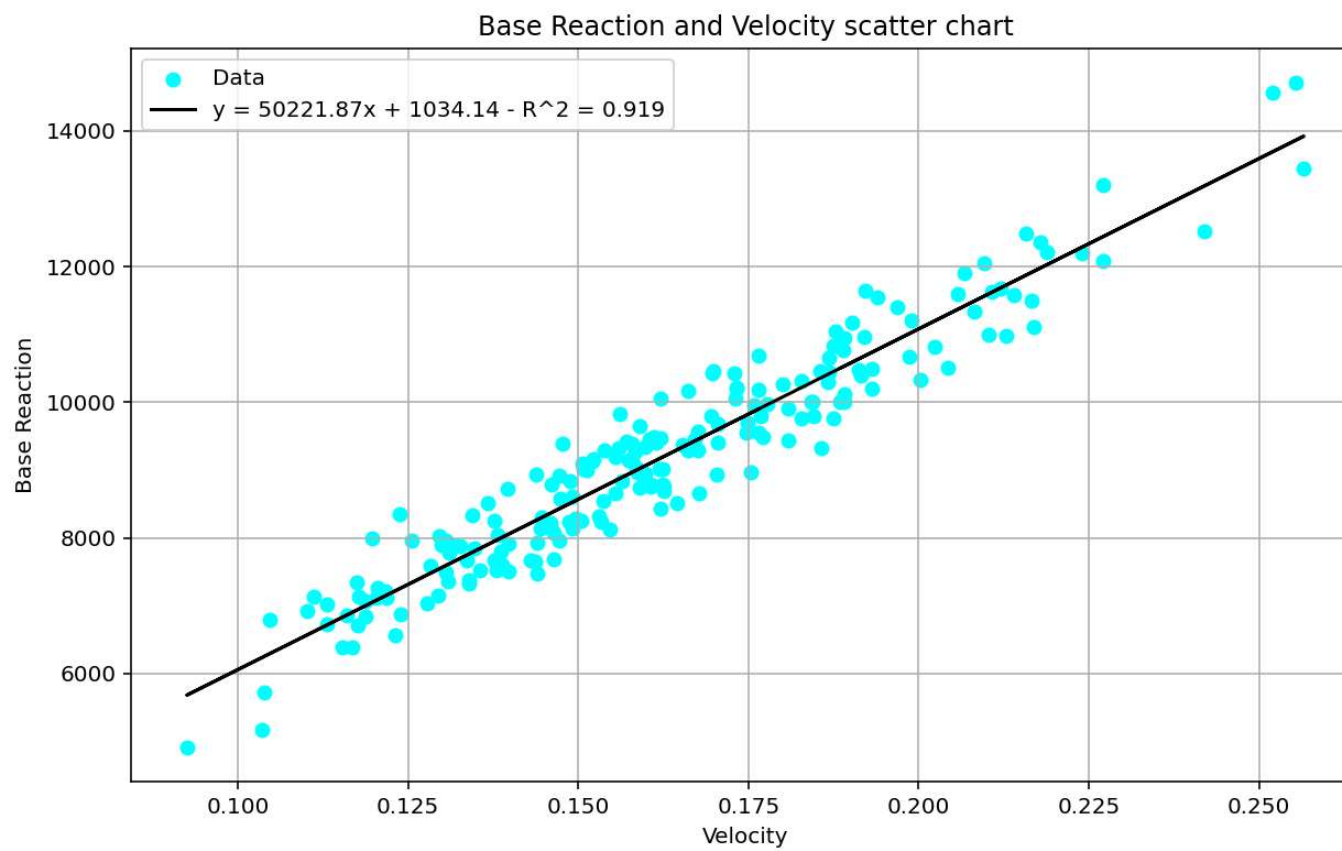


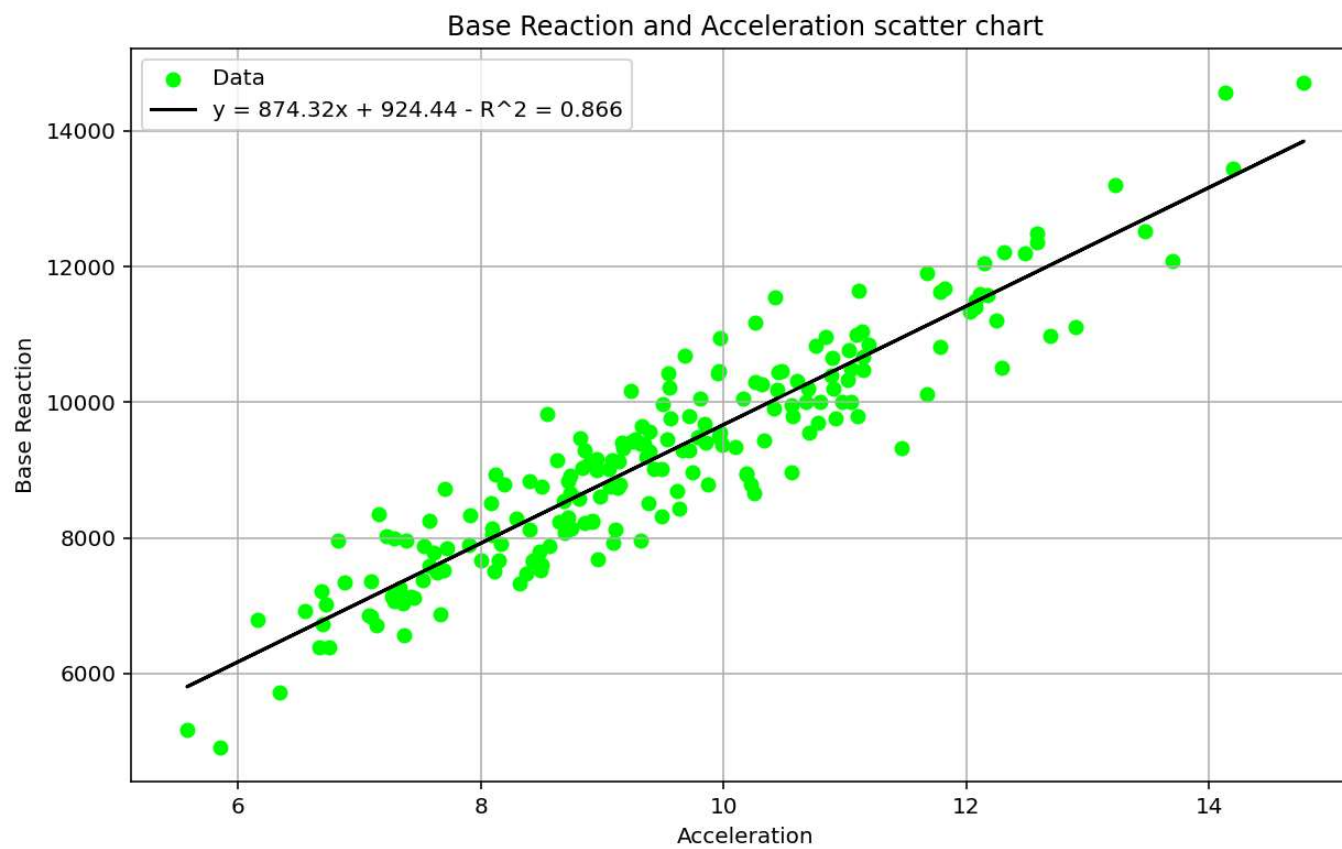


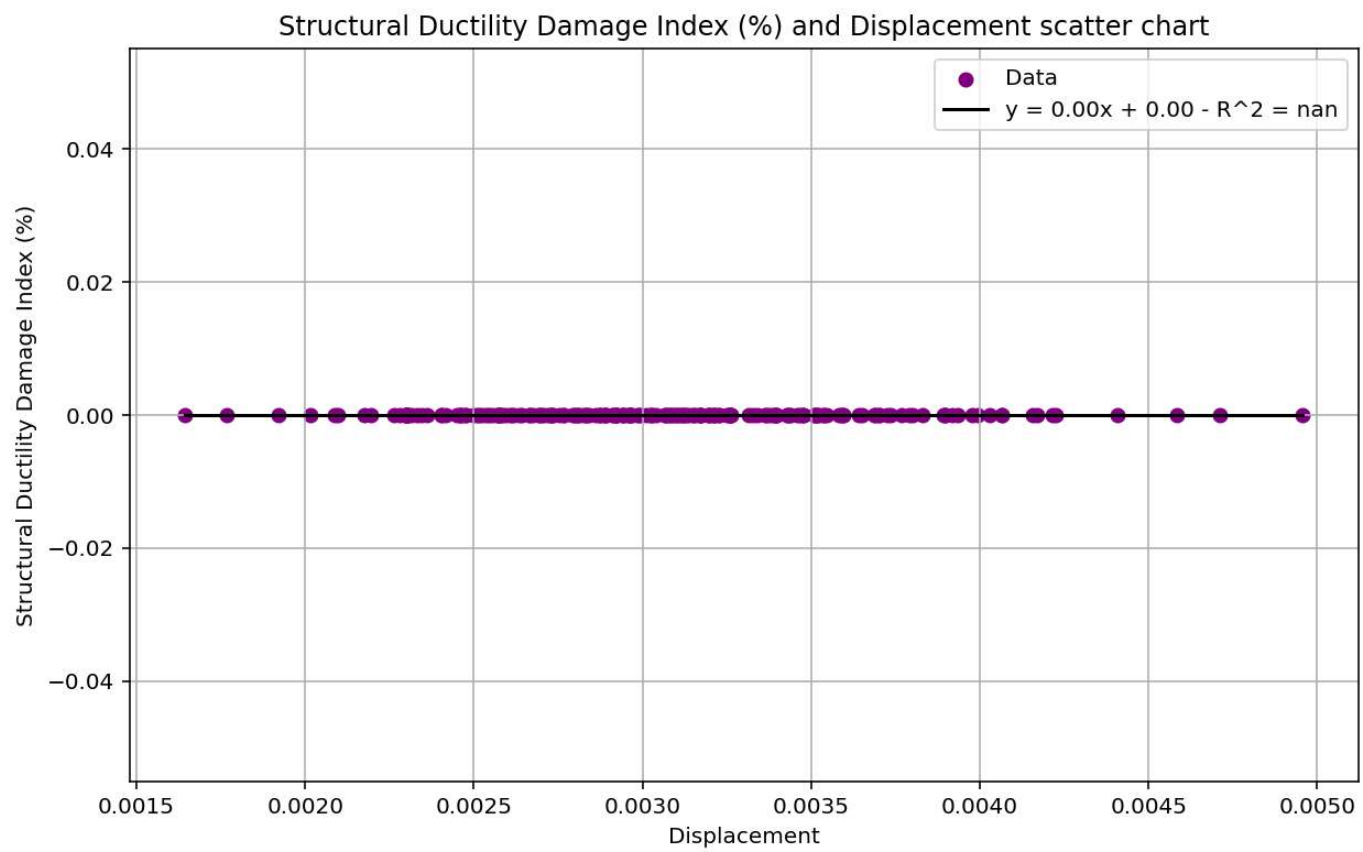


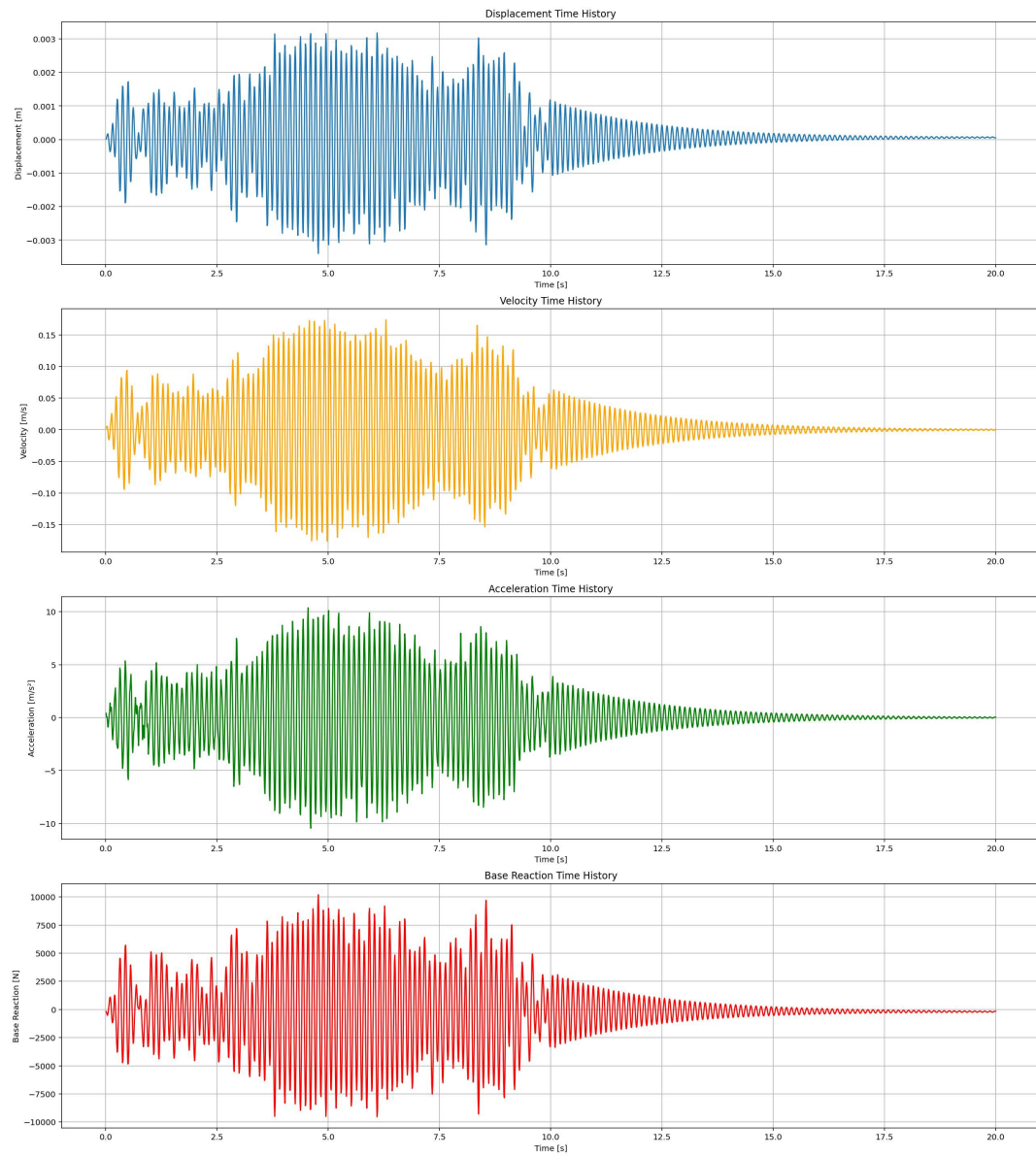
Base Reaction and Displacement scatter chart

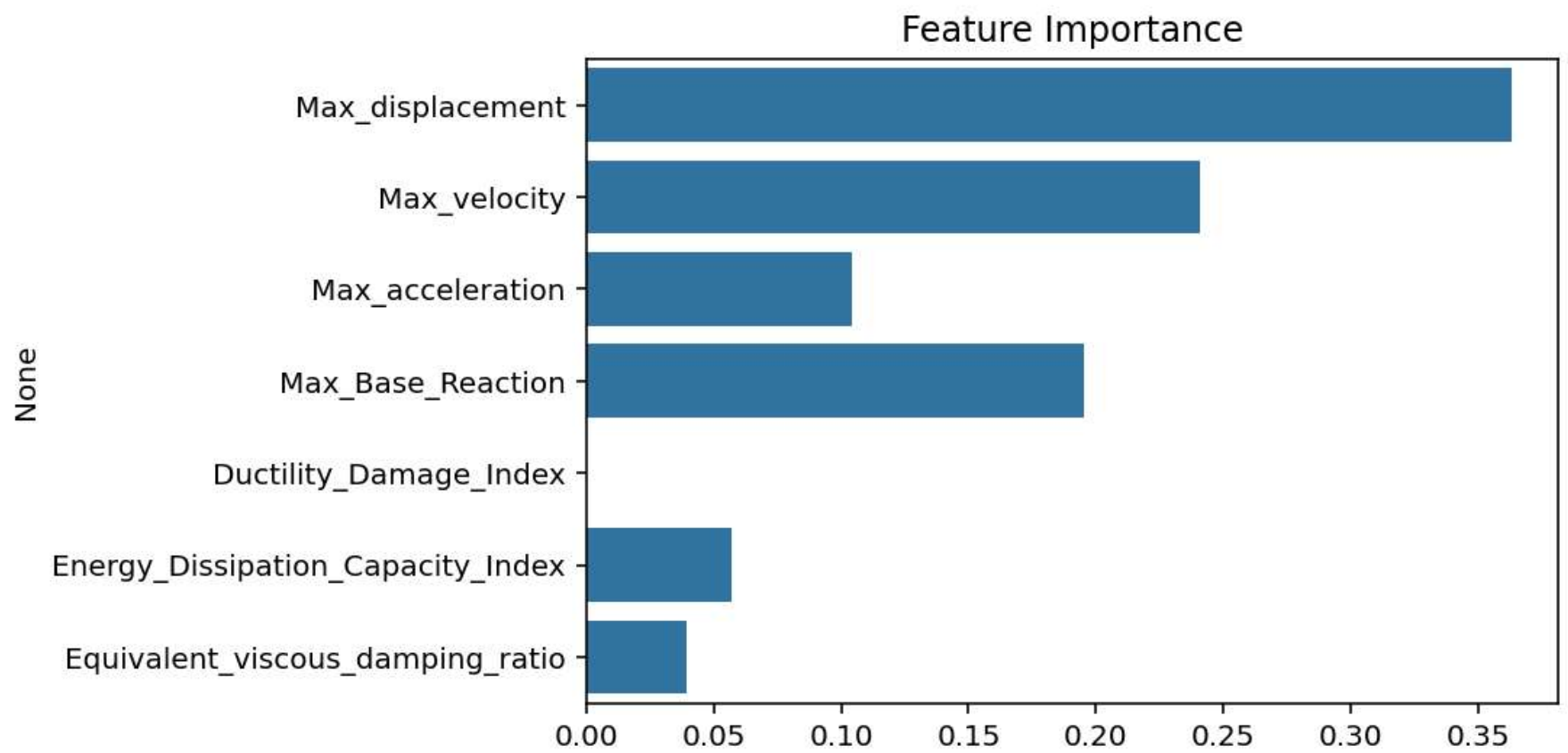












Correlation Heatmap

